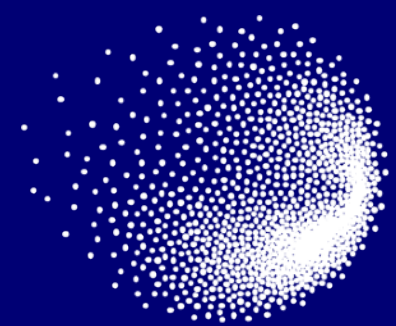
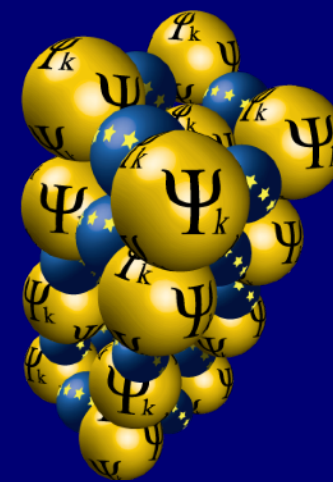




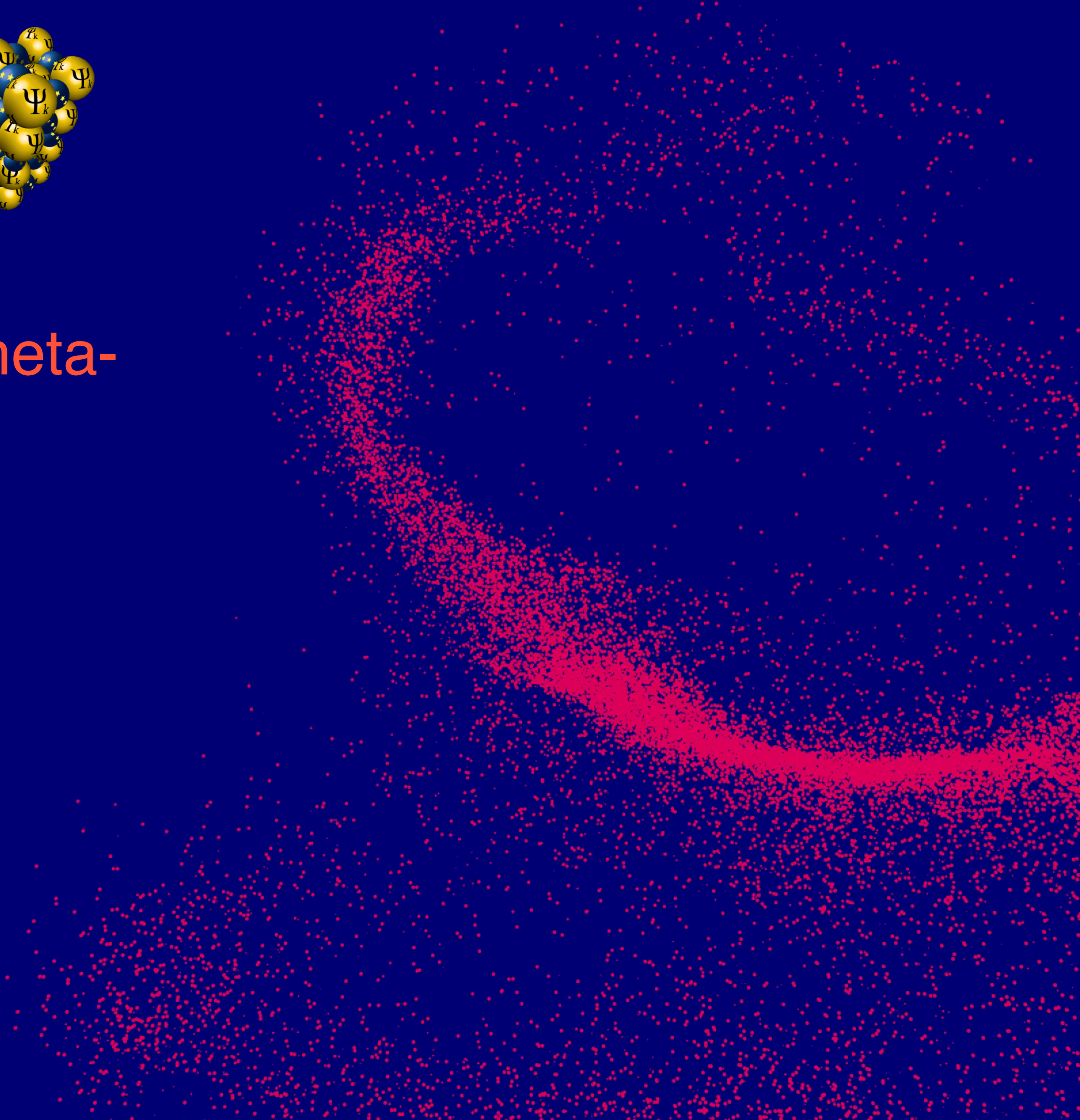
PSI ETH4D

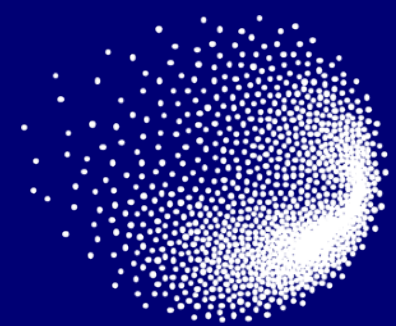


Hybrid, van der Waals, and meta-
GGA functionals

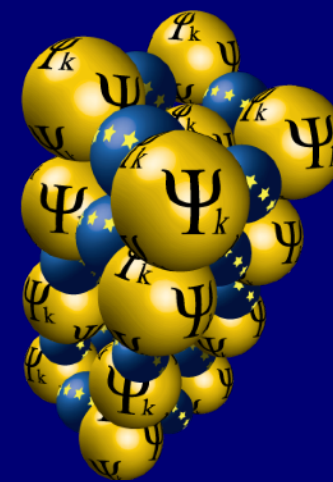
Iurii Timrov

Paul Scherrer Institut (PSI), Switzerland





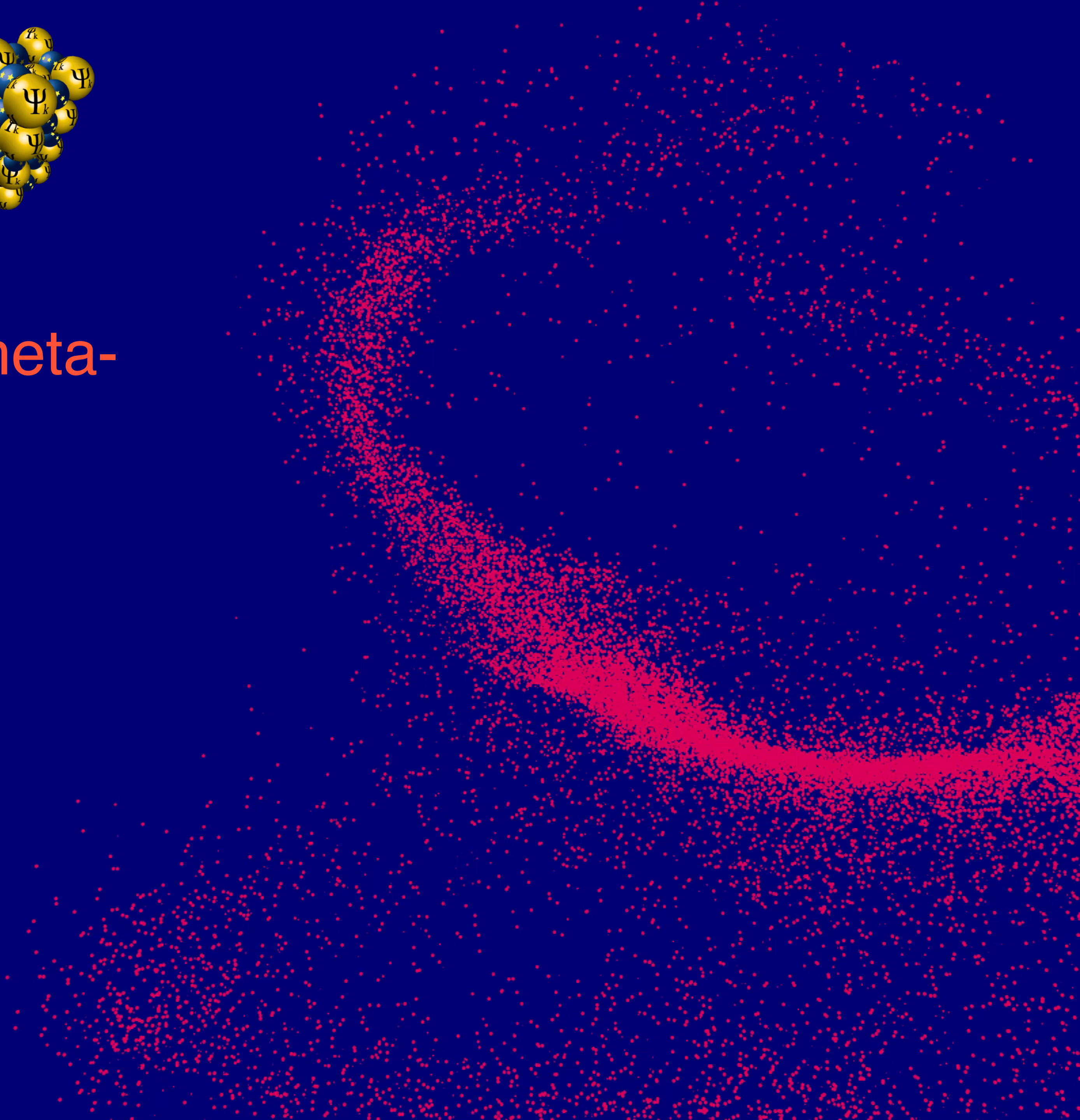
PSI ETH4D



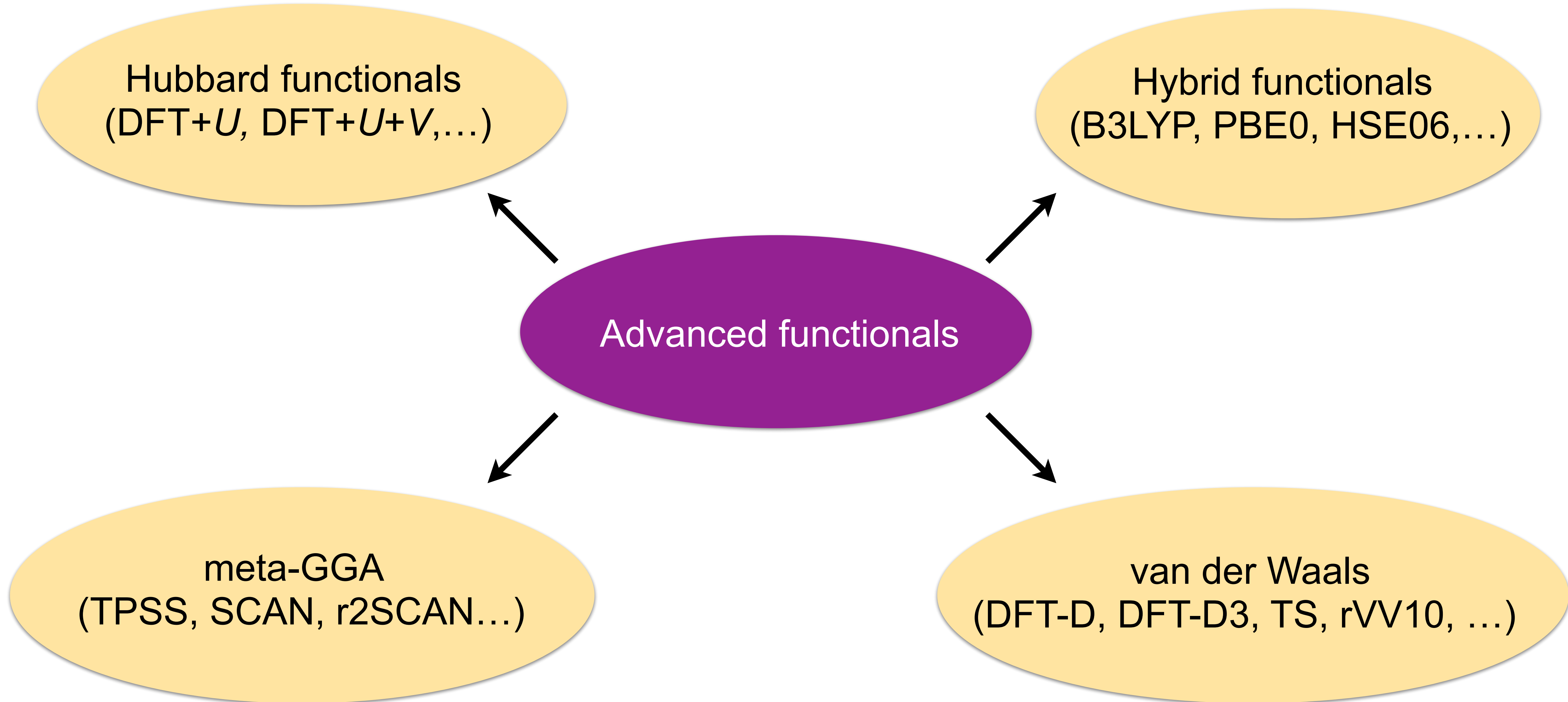
Hybrid, van der Waals, and meta-
GGA functionals

Iurii Timrov

Paul Scherrer Institut (PSI), Switzerland



Some advanced exchange-correlation functionals



Hybrid functionals

Exact (Fock) exchange

$$E_{\text{x}}^{\text{Fock}} = -\frac{1}{2} \sum_{\mathbf{k}} \sum_{\mathbf{k}'} \sum_v \sum_{v'} \iint \frac{\psi_{v,\mathbf{k}}^*(\mathbf{r}) \psi_{v',\mathbf{k}'}(\mathbf{r}) \psi_{v',\mathbf{k}'}^*(\mathbf{r}') \psi_{v,\mathbf{k}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'$$

double sum
over $\mathbf{k}$ points

double sum
over bands

$E_{\text{x}}^{\text{Fock}}$ is computationally very expensive in the plane-wave basis set

PBE0 hybrid functional

Mixing the exact exchange energy with the (semi-)local exchange-correlation functionals

$$E_{\text{xc}}^{\text{PBE0}} = \alpha E_{\text{x}}^{\text{Fock}} + (1 - \alpha) E_{\text{x}}^{\text{PBE}} + E_{\text{c}}^{\text{PBE}}$$

25% of the (non-local) exact-exchange (Fock) energy

75% of the (semi-local) exchange GGA-PBE energy

100% of the (semi-local) correlation GGA-PBE energy

PBE0 hybrid functional

Mixing the exact exchange energy with the (semi-)local exchange-correlation functionals

$$\alpha = 0.25$$

$$E_{\text{xc}}^{\text{PBE0}} = \alpha E_{\text{x}}^{\text{Fock}} + (1 - \alpha) E_{\text{x}}^{\text{PBE}} + E_{\text{c}}^{\text{PBE}}$$

25% of the (non-local) exact-exchange (Fock) energy

75% of the (semi-local) exchange GGA-PBE energy

100% of the (semi-local) correlation GGA-PBE energy

Molecules

Electronic atomization energies (i.e. energy required to completely separate a molecule into its constituent atoms) in kcal/mol

Molecule	PBE	PBE0	Expt.
N ₂	244	225	227
O ₂	143	124	118
SO	142	128	122

PBE0 hybrid functional gives much more accurate results than PBE compared to experiments on simple molecules

Solids

Solid	PBE		PBE0		Expt. [‡]	
	a_0	B_0	a_0	B_0	a_0	B_0
Li	3.438	13.7	3.463	13.7	3.477	13.0
Na	4.200	7.80	4.229	8.22	4.225	7.5
Al	4.040	76.6	4.012	86.0	4.032	79.4
BN	3.626	370	3.600	402	3.616	400
BP	4.547	160	4.520	174	4.538	165
C	3.574	431	3.549	467	3.567	443
Si	5.469	87.8	5.433	99.0	5.430	99.2
SiC	4.380	210	4.347	231	4.358	225
β -GaN	4.546	169	4.481	199	4.520	210
GaP	5.506	75.3	5.446	87.3	5.451	88.7
GaAs	5.752	59.9	5.682	72.7	5.648	75.6
LiF	4.068	67.3	4.011	72.8	4.010	69.8

PBE0 provides more accurate results than PBE for the lattice parameters and bulk modulus of solids

Solids

Direct and indirect band gaps

	PBE	PBE0	Expt.
GaAs			
$\Gamma_{15v} \rightarrow \Gamma_{1c}$	0.56	2.01	1.52 ^a
$\Gamma_{15v} \rightarrow X_{1c}$	1.46	2.67	1.90 ^a
$\Gamma_{15v} \rightarrow L_{1c}$	1.02	2.37	1.74 ^a
Si			
$\Gamma'_{25v} \rightarrow \Gamma_{15c}$	2.57	3.97	3.34–3.36, ^b 3.05 ^c
$\Gamma'_{25v} \rightarrow X_{1c}$	0.71	1.93	1.13, ^d 1.25 ^c
$\Gamma'_{25v} \rightarrow L_{1c}$	1.54	2.88	2.06(3), ^e 2.40(15) ^f
C			
$\Gamma'_{25v} \rightarrow \Gamma_{15}$	5.59	7.69	7.3 ^a
$\Gamma'_{25v} \rightarrow X_{1c}$	4.76	6.66	
$\Gamma'_{25v} \rightarrow L_{1c}$	8.46	10.77	
MgO			
$\Gamma_{15} \rightarrow \Gamma_1$	4.75	7.24	7.7 ^g
$X_{4'} \rightarrow \Gamma_1$	9.15	11.67	
$L_1 \rightarrow \Gamma_1$	7.91	10.38	

- PBE underestimates band gaps
- PBE0 predicts more accurate band gaps, but this is not systematic and there are overestimations
- Overestimation of band gaps in PBE0 is related to the non-optimal fraction of the Fock exchange (see next slides)

B3LYP hybrid functional

$$\alpha = 0.20, \beta = 0.72, \gamma = 0.81$$

$$E_{\text{xc}}^{\text{B3LYP}} = \alpha E_{\text{x}}^{\text{Fock}} + (1 - \alpha) E_{\text{x}}^{\text{LDA}} + \beta \Delta E_{\text{x}}^{\text{B88}} + \gamma E_{\text{c}}^{\text{LYP}} + (1 - \gamma) E_{\text{c}}^{\text{LDA}}$$

20% of the (non-local) exact-exchange (Fock) energy

80% of the (local) exchange LDA energy

Becke's gradient correction to the exchange functional

81% of the (local) correlation GGA-LYP energy

19% of the (local) correlation LDA energy

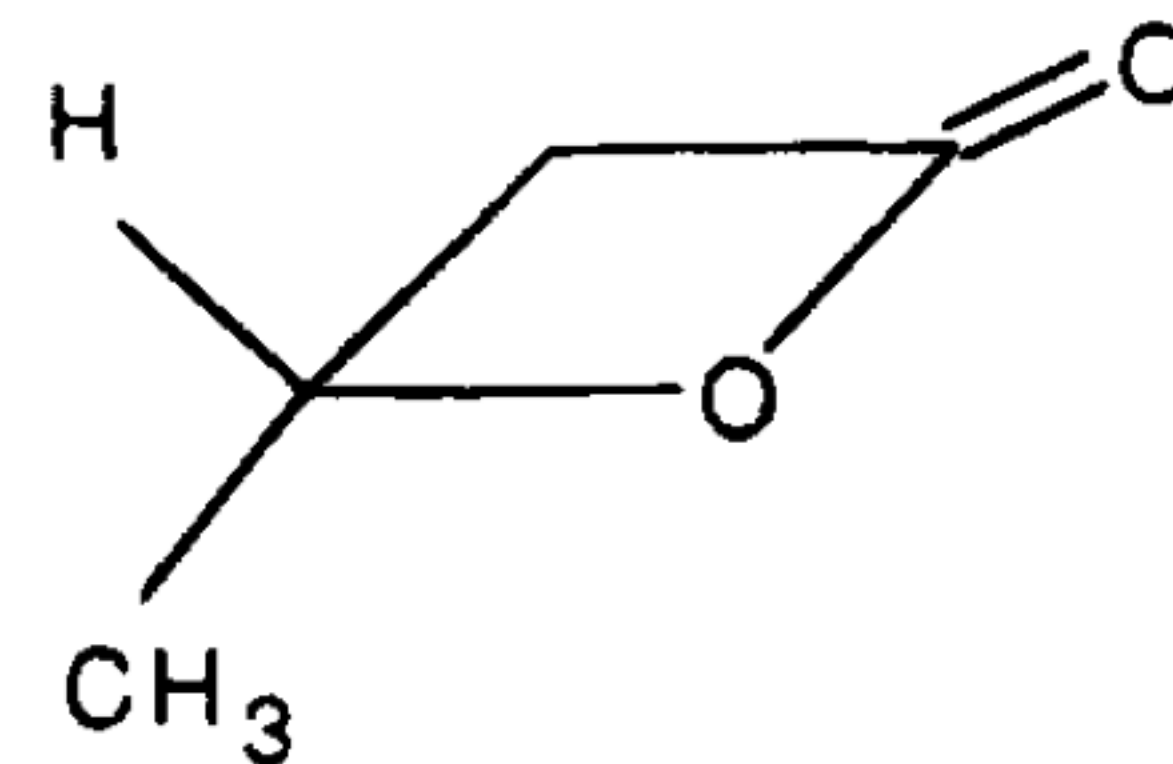
A.D. Becke, JCP 98, 1372 (1993).

P.J. Stephens, F.J. Devlin, C.F. Chabalowski, M.J. Frisch, JPC 98, 11623 (1994).

Molecules

Frequencies (ν), dipole strengths (D), rotational strengths (R) of 4-methyl-2-oxetanone

MP2			DFT/B3LYP			expt ^b		
ν	D	R	ν	D	R	ν	D	R
1528	14	-5	1500	13	-5	1453	25	-2
1516	8	4	1489	8	5	1441	3	3
1479	47	-11	1462	43	-11	1419	35	-9
1438	52	11	1422	51	11	1387	58	13
1394	52	27	1389	50	28	1350	40	42
1320	97	9	1316	112	3	1284	161	26
1230	47	-11	1229	44	-4	1198	47	-21
1216	14	0	1204	19	-5	1178	35	-17
1146	191	82	1137	327	99	1118	483	178
1125	219	-40	1113	87	-58	1099	51	-83
1077	10	-12	1076	11	-6	1055	40	-23
1056	271	-50	1041	268	-50	1022	251	-102
973	126	10	966	91	7	959	70	12
914	9	-29	914	8	-27	896	10	-54
873	221	33	850	310	46	836	371	78
829	97	26	817	34	10	812	55	29
714	3	-3	712	2	2	711	13	15



MP2 = second-order Møller-Plesset
perturbation theory

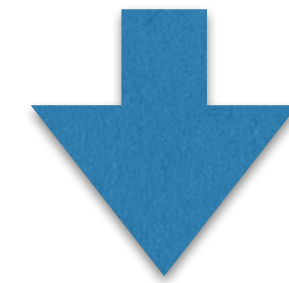
B3LYP is very popular in the quantum
chemistry community (study of molecules)

^a Frequencies, ν , in cm^{-1} , dipole strengths D in $10^{-40} \text{ esu}^2 \text{ cm}^2$, and rotational strengths R in $10^{-44} \text{ esu}^2 \text{ cm}^2$. R values are for $(R)-(+)-1$.

Range separation

The Coulomb potential is partitioned as a sum of the **short-range (SR)** and **long-range (LR)** parts:

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{\text{erfc}(\omega|\mathbf{r} - \mathbf{r}'|)}{|\mathbf{r} - \mathbf{r}'|} + \frac{\text{erf}(\omega|\mathbf{r} - \mathbf{r}'|)}{|\mathbf{r} - \mathbf{r}'|}$$

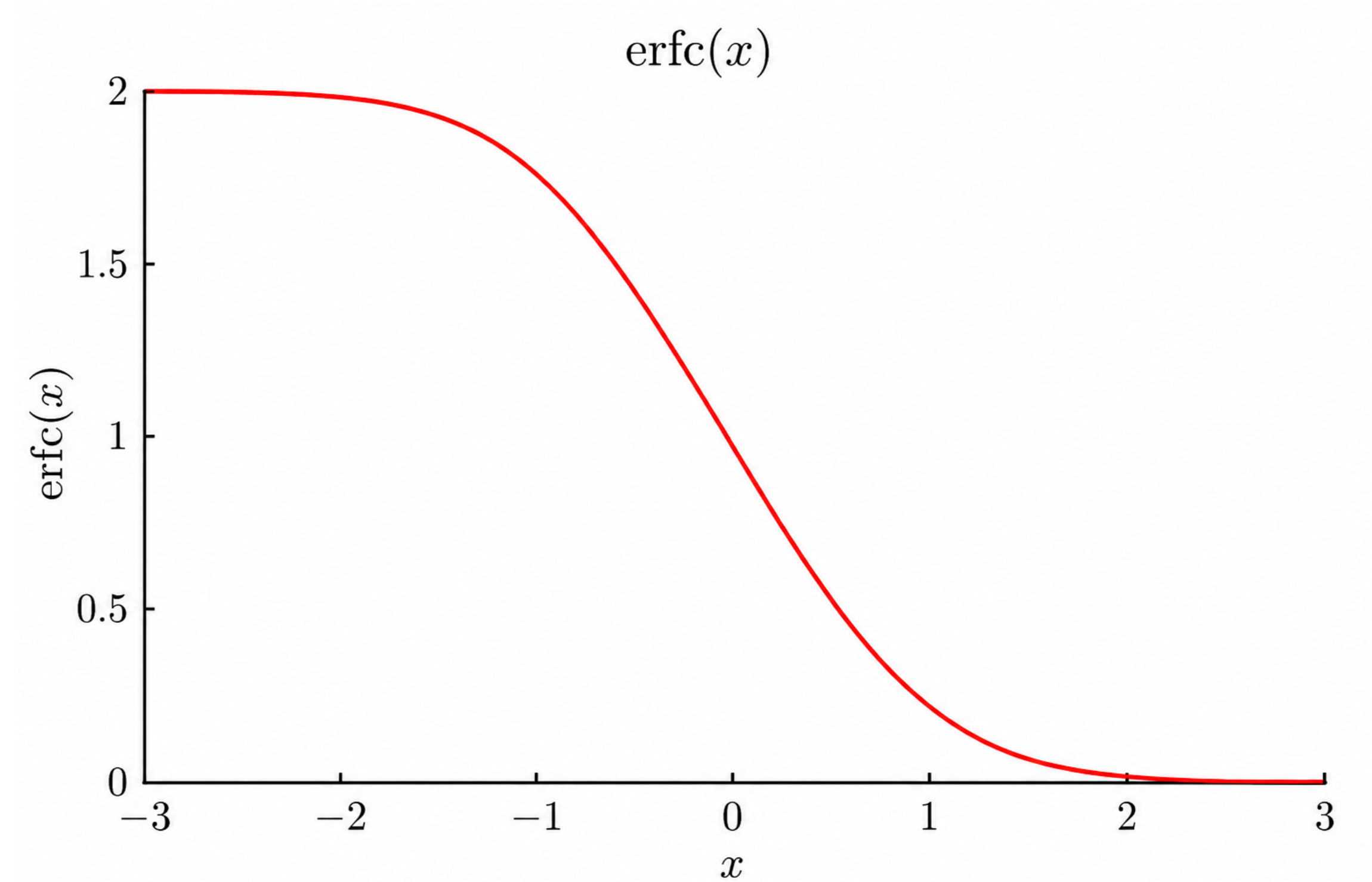
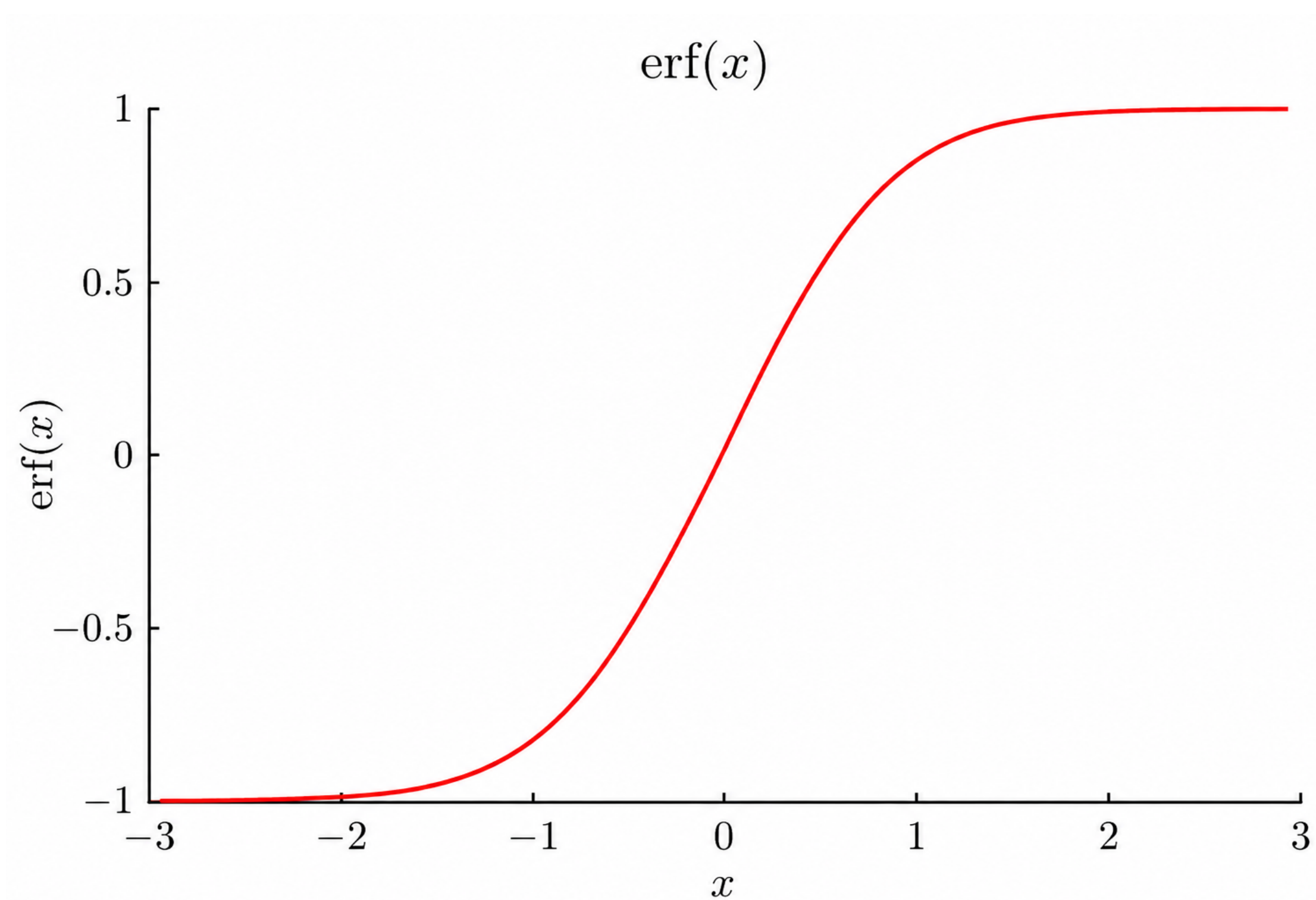


Range-separated hybrid functional

Error function and complementary error function

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-y^2} dy$$

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-y^2} dy$$



From PBE0 to HSE hybrid functional

$$\alpha = 0.25$$

$$E_{\text{xc}}^{\text{PBE0}} = \alpha E_{\text{x}}^{\text{Fock}} + (1 - \alpha) E_{\text{x}}^{\text{PBE}} + E_{\text{c}}^{\text{PBE}}$$

$$E_{\text{xc}}^{\text{HSE06}} = \alpha E_{\text{x}}^{\text{Fock,SR}} + (1 - \alpha) E_{\text{x}}^{\text{PBE,SR}} + E_{\text{x}}^{\text{PBE,LR}} + E_{\text{c}}^{\text{PBE}}$$

Solids

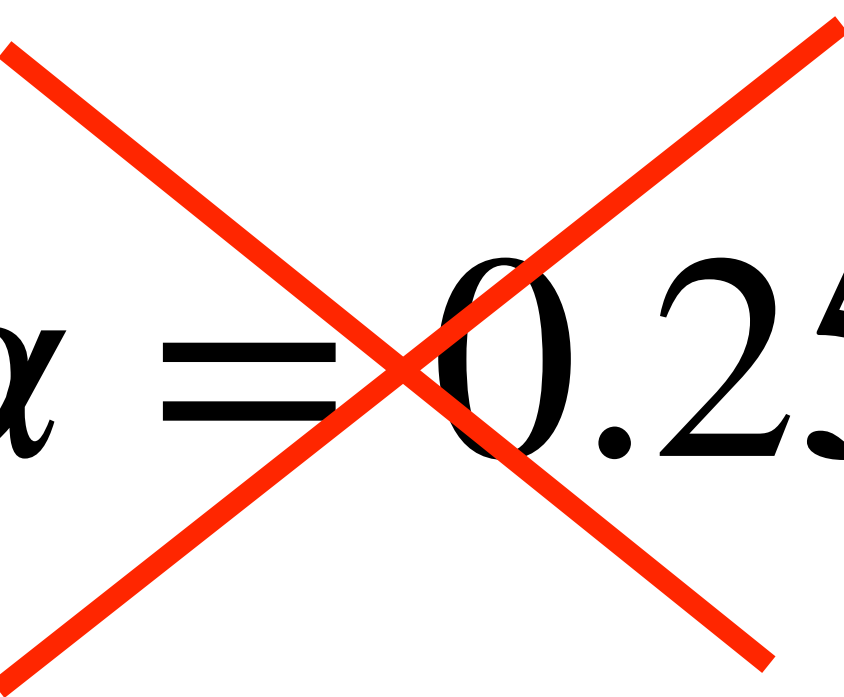
Direct and indirect band gaps

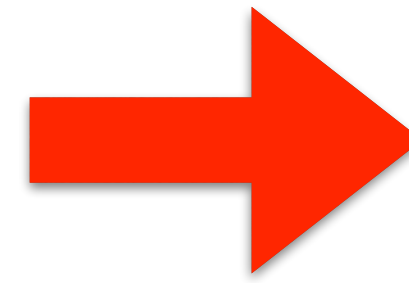
	PBE	PBE0	HSE03	Expt.
GaAs				
$\Gamma_{15v} \rightarrow \Gamma_{1c}$	0.56	2.01	1.29	1.52 ^a
$\Gamma_{15v} \rightarrow X_{1c}$	1.46	2.67	1.99	1.90 ^a
$\Gamma_{15v} \rightarrow L_{1c}$	1.02	2.37	1.64	1.74 ^a
Si				
$\Gamma'_{25v} \rightarrow \Gamma_{15c}$	2.57	3.97	3.19	3.34–3.36, ^b 3.05 ^c
$\Gamma'_{25v} \rightarrow X_{1c}$	0.71	1.93	1.23	1.13, ^d 1.25 ^c
$\Gamma'_{25v} \rightarrow L_{1c}$	1.54	2.88	2.13	2.06(3), ^e 2.40(15) ^f
C				
$\Gamma'_{25v} \rightarrow \Gamma_{15}$	5.59	7.69	6.77	7.3 ^a
$\Gamma'_{25v} \rightarrow X_{1c}$	4.76	6.66	5.76	
$\Gamma'_{25v} \rightarrow L_{1c}$	8.46	10.77	9.80	
MgO				
$\Gamma_{15} \rightarrow \Gamma_1$	4.75	7.24	6.34	7.7 ^g
$X_{4'} \rightarrow \Gamma_1$	9.15	11.67	10.79	
$L_1 \rightarrow \Gamma_1$	7.91	10.38	9.49	

J. Paier, M. Marsman, K. Hummer, G. Kresse, I.C. Gerber, J.G. Angyan, JCP 124, 154709 (2006).

What fraction of Fock exchange to use in solids (α parameter)?

From the analogy with the many-body perturbation theory, one can realize that:

$$\alpha = 0.25$$




$$\alpha = \frac{1}{\epsilon_{\infty}}$$

ϵ_{∞} is the electronic dielectric constant of a material

Dielectric constant ϵ_{∞} of solids

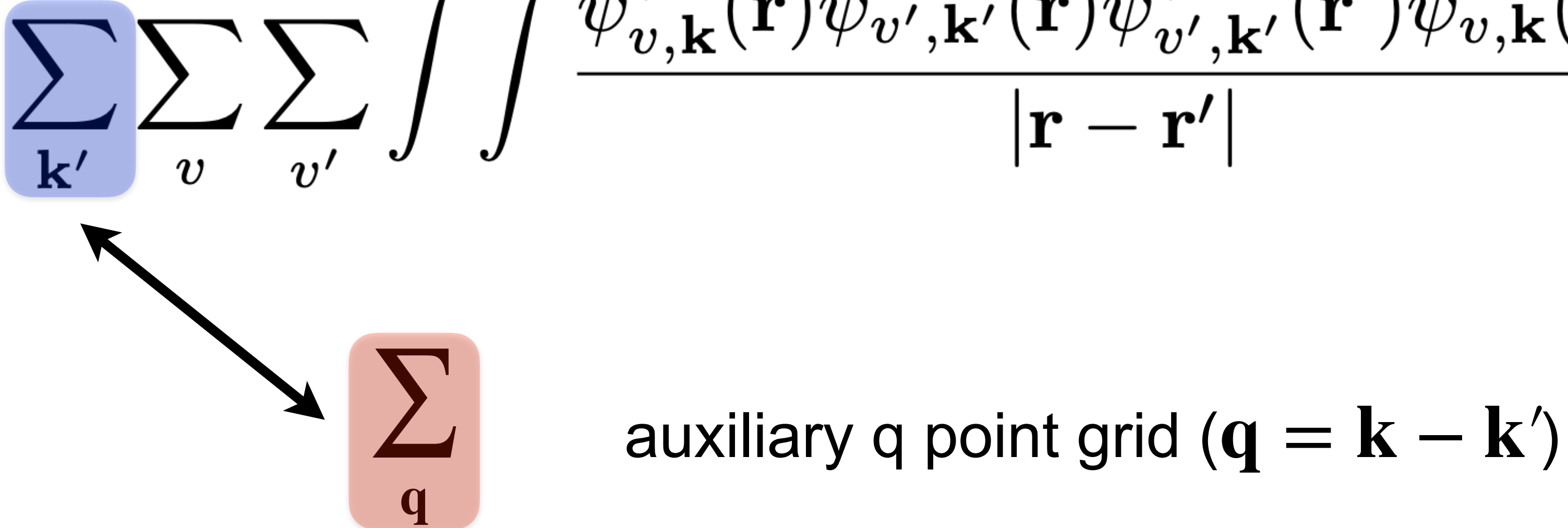
Material	ϵ_{∞}
Ge	15.9
Si	11.9
AlP	7.54
SiC	6.52
TiO ₂	6.34
NiO	5.76
C	5.70
CoO	5.35
GaN	5.30
ZnS	5.13
MnO	4.95
WO ₃	4.81

Material	ϵ_{∞}
BN	4.50
HfO ₂	4.41
AlN	4.18
ZnO	3.74
Al ₂ O ₃	3.10
MgO	2.96
LiCl	2.70
NaCl	2.40
LiF	1.90
H ₂ O	1.72
Ar	1.66
Ne	1.23

Band gaps of solids

Material	PBE	PBE0 ($\alpha = 0.25$)	PBE0 modified ($\alpha = 1/\epsilon_\infty$)	Expt.
Ge	0.00	1.53	0.77	0.74
Si	0.62	1.75	1.03	1.17
AlP	1.64	2.98	2.41	2.51
SiC	1.37	2.91	2.33	2.39
TiO ₂	1.81	3.92	3.18	3.3
NiO	0.97	5.28	4.61	4.3
C	4.15	5.95	5.44	5.48
CoO	0.00	4.53	4.01	2.5
GaN	1.88	3.68	3.30	3.29
ZnS	2.36	4.18	3.85	3.91
MnO	1.12	3.87	3.66	3.9
WO ₃	1.92	3.79	3.50	3.38

Calculation of the exact energy in practice

$$E_{\text{x}}^{\text{Fock}} = -\frac{1}{2} \sum_{\mathbf{k}} \sum_{\mathbf{k}'} \sum_v \sum_{v'} \iint \frac{\psi_{v,\mathbf{k}}^*(\mathbf{r}) \psi_{v',\mathbf{k}'}(\mathbf{r}) \psi_{v',\mathbf{k}'}^*(\mathbf{r}') \psi_{v,\mathbf{k}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'$$


auxiliary q point grid ($\mathbf{q} = \mathbf{k} - \mathbf{k}'$)

Replace one sum over k points with a sum over q points ($N_{\mathbf{q}} \leq N_{\mathbf{k}}$)

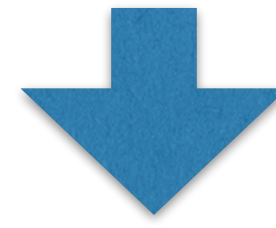
This allows us to speed up significantly the calculation of $E_{\text{x}}^{\text{Fock}}$

From real to reciprocal space

$$E_{\text{x}}^{\text{Fock}} = -\frac{1}{2} \sum_{\mathbf{k}} \sum_{\mathbf{k}'} \sum_v \sum_{v'} \iint \frac{\psi_{v,\mathbf{k}}^*(\mathbf{r}) \psi_{v',\mathbf{k}'}(\mathbf{r}) \psi_{v',\mathbf{k}'}^*(\mathbf{r}') \psi_{v,\mathbf{k}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'$$

From real to reciprocal space

$$E_{\text{x}}^{\text{Fock}} = -\frac{1}{2} \sum_{\mathbf{k}} \sum_{\mathbf{k}'} \sum_v \sum_{v'} \iint \frac{\psi_{v,\mathbf{k}}^*(\mathbf{r}) \psi_{v',\mathbf{k}'}(\mathbf{r}) \psi_{v',\mathbf{k}'}^*(\mathbf{r}') \psi_{v,\mathbf{k}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'$$



$$E_{\text{x}}^{\text{Fock}} = -\frac{2\pi}{\Omega} \frac{1}{N_{\mathbf{q}}} \sum_{\mathbf{q}} \sum_{\mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G})}{|\mathbf{q} + \mathbf{G}|^2}, \quad A(\mathbf{q} + \mathbf{G}) = \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{k}} \sum_{v,v'} |\rho_{v,\mathbf{k}}^{v',\mathbf{k}-\mathbf{q}}(\mathbf{q} + \mathbf{G})|^2$$

From real to reciprocal space

$$E_{\text{x}}^{\text{Fock}} = -\frac{1}{2} \sum_{\mathbf{k}} \sum_{\mathbf{k}'} \sum_v \sum_{v'} \iint \frac{\psi_{v,\mathbf{k}}^*(\mathbf{r}) \psi_{v',\mathbf{k}'}(\mathbf{r}) \psi_{v',\mathbf{k}'}^*(\mathbf{r}') \psi_{v,\mathbf{k}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'$$



$$E_{\text{x}}^{\text{Fock}} = -\frac{2\pi}{\Omega} \frac{1}{N_{\mathbf{q}}} \sum_{\mathbf{q}} \sum_{\mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G})}{|\mathbf{q} + \mathbf{G}|^2}, \quad A(\mathbf{q} + \mathbf{G}) = \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{k}} \sum_{v,v'} |\rho_{v,\mathbf{k}}^{v',\mathbf{k}-\mathbf{q}}(\mathbf{q} + \mathbf{G})|^2$$

$$\rho_{v,\mathbf{k}}^{v',\mathbf{k}-\mathbf{q}}(\mathbf{r}) = u_{v',\mathbf{k}-\mathbf{q}}^*(\mathbf{r}) u_{v,\mathbf{k}}(\mathbf{r})$$

From real to reciprocal space

$$E_{\text{x}}^{\text{Fock}} = -\frac{1}{2} \sum_{\mathbf{k}} \sum_{\mathbf{k}'} \sum_v \sum_{v'} \iint \frac{\psi_{v,\mathbf{k}}^*(\mathbf{r}) \psi_{v',\mathbf{k}'}(\mathbf{r}) \psi_{v',\mathbf{k}'}^*(\mathbf{r}') \psi_{v,\mathbf{k}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'$$



$$E_{\text{x}}^{\text{Fock}} = -\frac{2\pi}{\Omega} \frac{1}{N_{\mathbf{q}}} \sum_{\mathbf{q}} \sum_{\mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G})}{|\mathbf{q} + \mathbf{G}|^2},$$

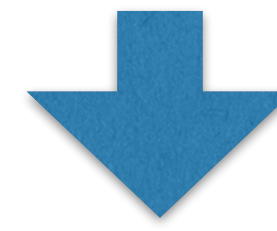
$$A(\mathbf{q} + \mathbf{G}) = \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{k}} \sum_{v,v'} |\rho_{v,\mathbf{k}}^{v',\mathbf{k}-\mathbf{q}}(\mathbf{q} + \mathbf{G})|^2$$

$$\rho_{v,\mathbf{k}}^{v',\mathbf{k}-\mathbf{q}}(\mathbf{r}) = u_{v',\mathbf{k}-\mathbf{q}}^*(\mathbf{r}) u_{v,\mathbf{k}}(\mathbf{r})$$

$$\psi_{v,\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N_{\mathbf{k}}}} e^{i\mathbf{k} \cdot \mathbf{r}} u_{v,\mathbf{k}}(\mathbf{r})$$

From real to reciprocal space

$$E_{\text{x}}^{\text{Fock}} = -\frac{1}{2} \sum_{\mathbf{k}} \sum_{\mathbf{k}'} \sum_v \sum_{v'} \iint \frac{\psi_{v,\mathbf{k}}^*(\mathbf{r}) \psi_{v',\mathbf{k}'}(\mathbf{r}) \psi_{v',\mathbf{k}'}^*(\mathbf{r}') \psi_{v,\mathbf{k}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'$$



$$E_{\text{x}}^{\text{Fock}} = -\frac{2\pi}{\Omega} \frac{1}{N_{\mathbf{q}}} \sum_{\mathbf{q}} \sum_{\mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G})}{|\mathbf{q} + \mathbf{G}|^2},$$

$$A(\mathbf{q} + \mathbf{G}) = \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{k}} \sum_{v,v'} |\rho_{v,\mathbf{k}}^{v',\mathbf{k}-\mathbf{q}}(\mathbf{q} + \mathbf{G})|^2$$

$$\rho_{v,\mathbf{k}}^{v',\mathbf{k}-\mathbf{q}}(\mathbf{r}) = u_{v',\mathbf{k}-\mathbf{q}}^*(\mathbf{r}) u_{v,\mathbf{k}}(\mathbf{r})$$

singularity when
 $\mathbf{q} + \mathbf{G} = 0$

$$\psi_{v,\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N_{\mathbf{k}}}} e^{i\mathbf{k} \cdot \mathbf{r}} u_{v,\mathbf{k}}(\mathbf{r})$$

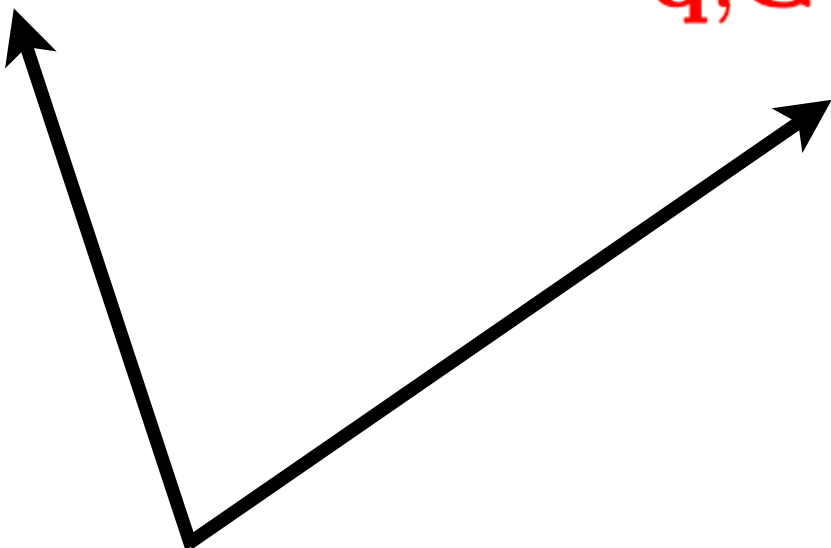
Gygi-Baldereschi method to treat the $\mathbf{q}+\mathbf{G}=0$ singularity

$$E_{\mathbf{x}}^{\text{Fock}} = -\frac{2\pi}{\Omega} \frac{1}{N_{\mathbf{q}}} \sum_{\mathbf{q}} \sum_{\mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G})}{|\mathbf{q} + \mathbf{G}|^2}$$

Gygi-Baldereschi method to treat the $q+G=0$ singularity

$$\begin{aligned} E_x^{\text{Fock}} &= -\frac{2\pi}{\Omega} \frac{1}{N_q} \sum_{\mathbf{q}} \sum_{\mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G})}{|\mathbf{q} + \mathbf{G}|^2} \\ &= -\frac{2\pi}{\Omega} \left\{ \frac{1}{N_q} \sum_{\mathbf{q}, \mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G}) - A(0)e^{-\alpha|\mathbf{q} + \mathbf{G}|^2}}{|\mathbf{q} + \mathbf{G}|^2} + \frac{1}{N_q} \sum_{\mathbf{q}, \mathbf{G}} \frac{e^{-\alpha|\mathbf{q} + \mathbf{G}|^2}}{|\mathbf{q} + \mathbf{G}|^2} A(0) \right\} \end{aligned}$$

We are adding and subtracting exactly the same term (in red)



Gygi-Baldereschi method to treat the $q+G=0$ singularity

$$\begin{aligned}
 E_x^{\text{Fock}} &= -\frac{2\pi}{\Omega} \frac{1}{N_q} \sum_{\mathbf{q}} \sum_{\mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G})}{|\mathbf{q} + \mathbf{G}|^2} \\
 &= -\frac{2\pi}{\Omega} \left\{ \frac{1}{N_q} \sum_{\mathbf{q}, \mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G}) - A(0)e^{-\alpha|\mathbf{q} + \mathbf{G}|^2}}{|\mathbf{q} + \mathbf{G}|^2} + \frac{1}{N_q} \sum_{\mathbf{q}, \mathbf{G}} \frac{e^{-\alpha|\mathbf{q} + \mathbf{G}|^2}}{|\mathbf{q} + \mathbf{G}|^2} A(0) \right\} \\
 &= -\frac{2\pi}{\Omega} \left\{ \frac{1}{N_q} \left[\sum'_{\mathbf{q}, \mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G})}{|\mathbf{q} + \mathbf{G}|^2} + \lim_{\mathbf{q} \rightarrow 0} \frac{A(\mathbf{q}) - A(0)}{q^2} \right] + D \times A(0) \right\}
 \end{aligned}$$

$$D = \frac{1}{N_q} \left[-\sum'_{\mathbf{q}, \mathbf{G}} \frac{e^{-\alpha|\mathbf{q} + \mathbf{G}|^2}}{|\mathbf{q} + \mathbf{G}|^2} + \alpha \right] + \frac{\Omega}{(2\pi)^3} \sqrt{\frac{\pi}{\alpha}}$$

Gygi-Baldereschi method to treat the $\mathbf{q}+\mathbf{G}=0$ singularity

$$\begin{aligned}
 E_{\mathbf{x}}^{\text{Fock}} &= -\frac{2\pi}{\Omega} \frac{1}{N_{\mathbf{q}}} \sum_{\mathbf{q}} \sum_{\mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G})}{|\mathbf{q} + \mathbf{G}|^2} \\
 &= -\frac{2\pi}{\Omega} \left\{ \frac{1}{N_{\mathbf{q}}} \sum_{\mathbf{q}, \mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G}) - A(0)e^{-\alpha|\mathbf{q}+\mathbf{G}|^2}}{|\mathbf{q} + \mathbf{G}|^2} + \frac{1}{N_{\mathbf{q}}} \sum_{\mathbf{q}, \mathbf{G}} \frac{e^{-\alpha|\mathbf{q}+\mathbf{G}|^2}}{|\mathbf{q} + \mathbf{G}|^2} A(0) \right\} \\
 &= -\frac{2\pi}{\Omega} \left\{ \frac{1}{N_{\mathbf{q}}} \left[\sum'_{\mathbf{q}, \mathbf{G}} \frac{A(\mathbf{q} + \mathbf{G})}{|\mathbf{q} + \mathbf{G}|^2} + \lim_{\mathbf{q} \rightarrow 0} \frac{A(\mathbf{q}) - A(0)}{\mathbf{q}^2} \right] + D \times A(0) \right\}
 \end{aligned}$$

no $\mathbf{q} + \mathbf{G} = 0$ singularity,
straightforward to compute

non-analytic limit 0/0
can be dealt with using
different techniques
(e.g. extrapolation)

computed analytically

meta-GGA functionals

Main idea

$$E_{\text{xc}}[n_{\uparrow}, n_{\downarrow}] = \int d^3r \, n(\mathbf{r}) \, \varepsilon_{\text{xc}} \left(n_{\uparrow}(\mathbf{r}), n_{\downarrow}(\mathbf{r}), \nabla n_{\uparrow}(\mathbf{r}), \nabla n_{\downarrow}(\mathbf{r}), \nabla^2 n_{\uparrow}(\mathbf{r}), \nabla^2 n_{\downarrow}(\mathbf{r}), \tau_{\uparrow}(\mathbf{r}), \tau_{\downarrow}(\mathbf{r}) \right)$$

$$n(\mathbf{r}) = \sum_{\sigma} n_{\sigma}(\mathbf{r})$$

total charge density

$$n_{\sigma}(\mathbf{r}) = \sum_{v, \mathbf{k}} |\psi_{v, \mathbf{k}}^{\sigma}(\mathbf{r})|^2$$

spin-charge density

$$\sigma = \{\uparrow, \downarrow\}$$

$$\tau_{\sigma}(\mathbf{r}) = \frac{1}{2} \sum_{v, \mathbf{k}} |\nabla \psi_{v, \mathbf{k}}^{\sigma}(\mathbf{r})|^2$$

kinetic energy density

Different meta-GGA functionals

BR89

r²SCAN

MVS

TPSS

SCAN

rSCAN

MGGA_MS2

TASK

revTPSS

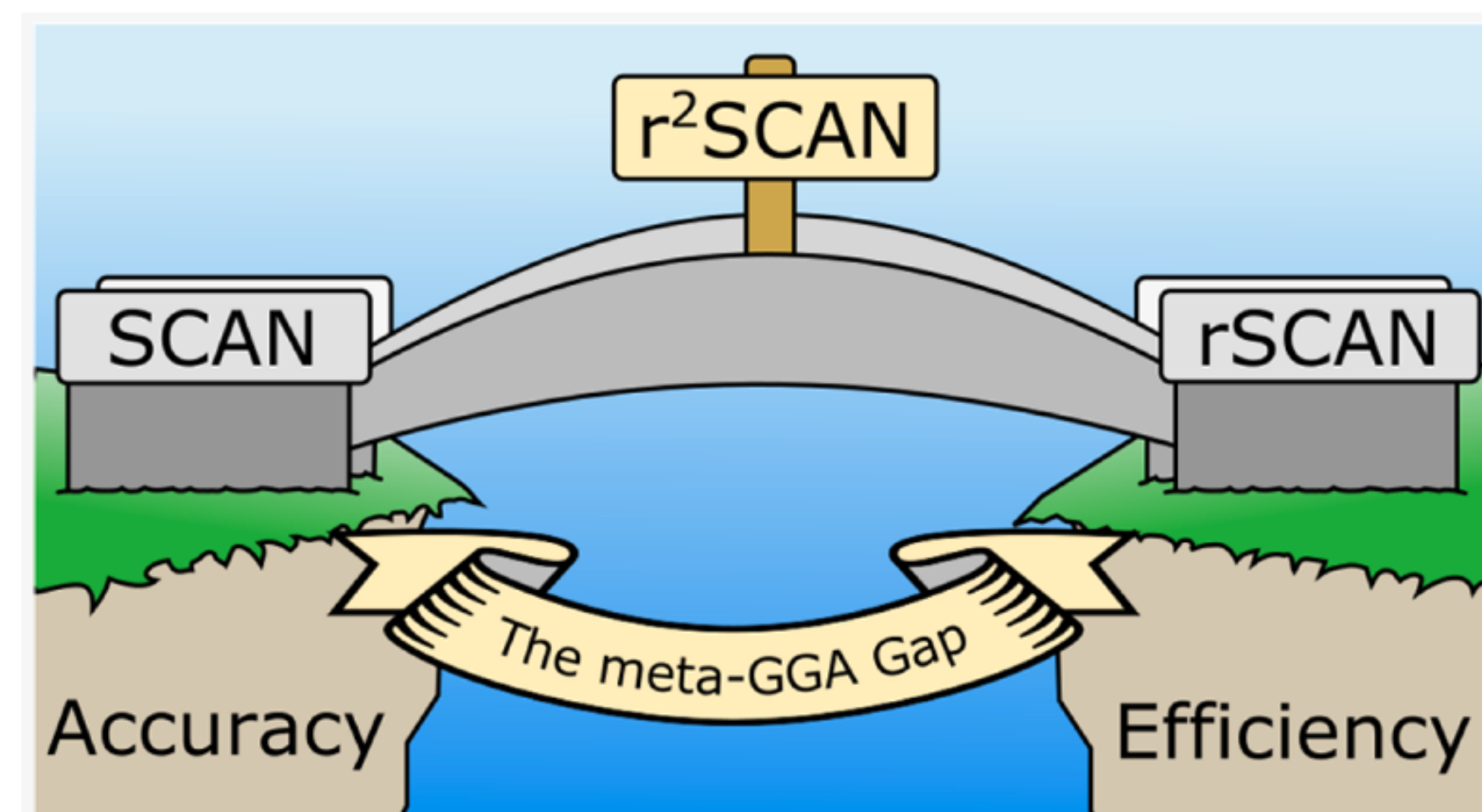
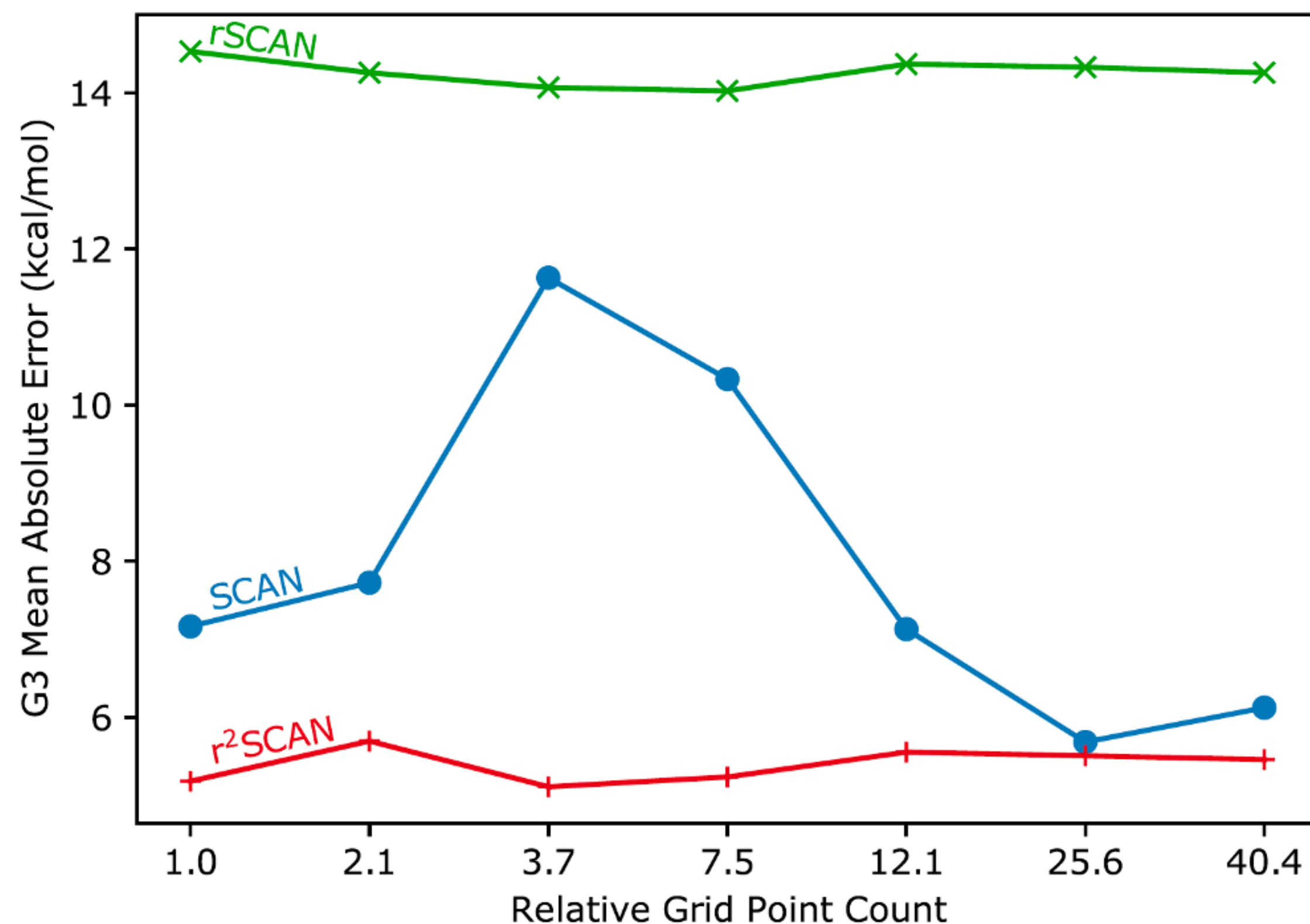
HLE17

TM

SCAN-L

SCAN, rSCAN, r2SCAN

$$E_{\text{xc}}[n_{\uparrow}, n_{\downarrow}] = \int d^3r n(\mathbf{r}) \varepsilon_{\text{xc}} \left(n_{\uparrow}(\mathbf{r}), n_{\downarrow}(\mathbf{r}), \nabla n_{\uparrow}(\mathbf{r}), \nabla n_{\downarrow}(\mathbf{r}), \cancel{\nabla^2 n_{\uparrow}(\mathbf{r})}, \cancel{\nabla^2 n_{\downarrow}(\mathbf{r})}, \tau_{\uparrow}(\mathbf{r}), \tau_{\downarrow}(\mathbf{r}) \right)$$



SCAN: J. Sun et al., *Phys. Rev. Lett.* **115**, 036402 (2015).

rSCAN: A. Bartok et al., *J. Chem. Phys.* **150**, 161101 (2019).

r2SCAN: J. Furness et al., *J. Phys. Chem. Lett.* **11**, 8208 (2020).

Accuracy of the SCAN meta-GGA functional

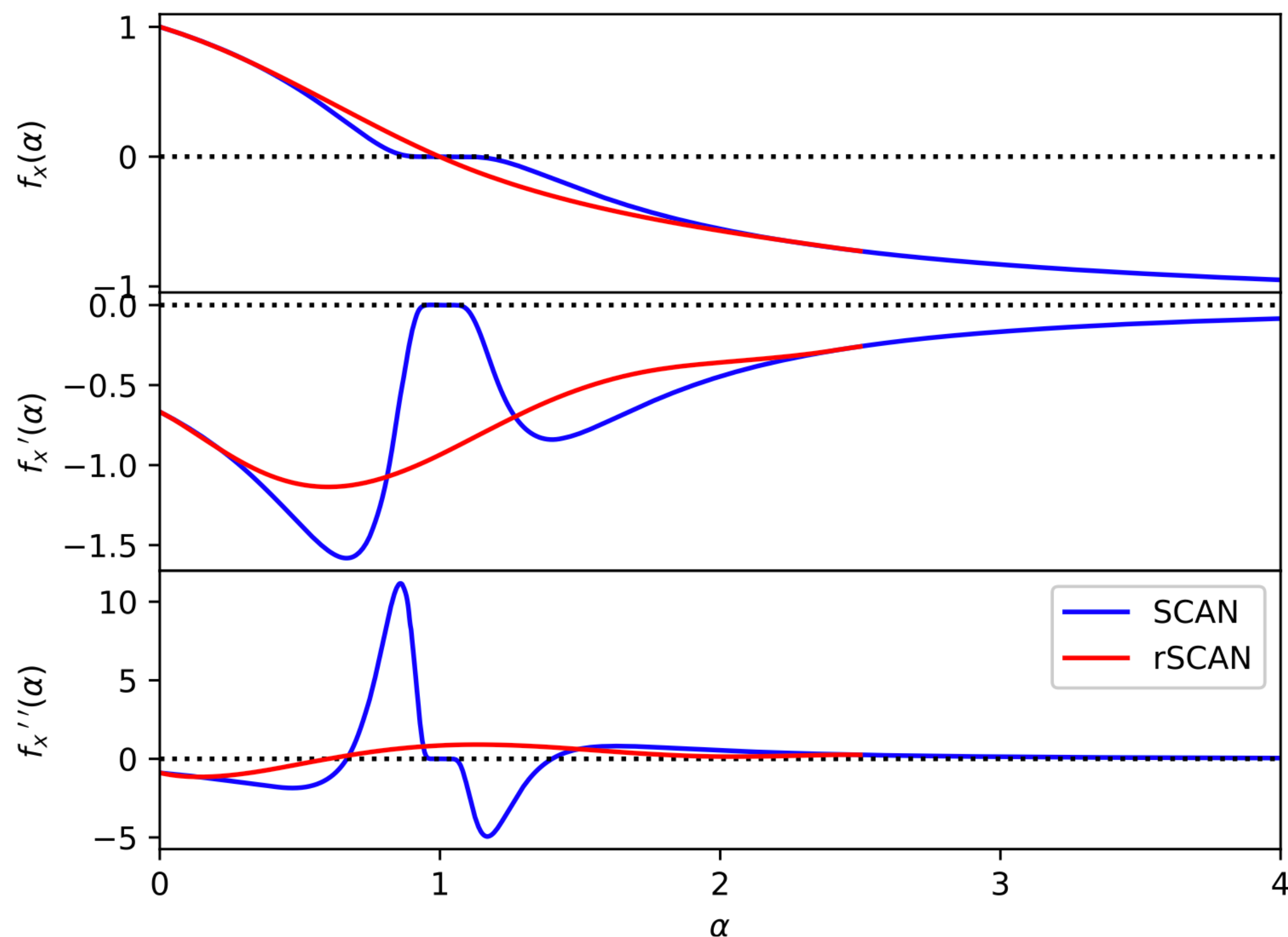
“SCAN” stands for Strongly Constrained and Appropriately Normed functional

SCAN obeys all 17 known exact constraints

	G 3 ^{HC}		G3		BH76		S22		LC20(Å)	
	ME	MAE	ME	MAE	ME	MAE	ME	MAE	ME	MAE
LSDA	−5.6	13.0	−83.7	83.7	−15.2	15.4	2.3	2.3	−0.081	0.081
BLYP	1.8	6.2	3.8	9.5	−7.9	7.9	−8.7	8.8		
PBEsol	−4.1	6.5	−58.7	58.8	−11.5	11.5	−1.3	1.8	−0.012	0.036
PBE	−2.1	6.6	−21.7	22.2	−9.1	9.2	−2.8	2.8	0.051	0.059
TPSS	1.9	3.8	−5.2	5.8	−8.6	8.7	−3.7	3.7	0.035	0.043
M06 L	−0.2	4.6	−1.6	5.2	−3.9	4.1	−0.9	0.9	0.015	0.069
SCAN	−0.8	2.7	−4.6	5.7	−7.7	7.7	−0.7	0.9	0.007	0.016

TABLE II. Mean error (ME) and mean absolute error (MAE) of SCAN and other semilocal functionals for the G3 set of molecules [73], the BH76 set of chemical barrier heights [74], the S22 set of weakly bonded complexes [69], and the LC20 set of solid lattice constants [75]. For the G3-1 subset of small molecules, the SCAN MAE is 3.2 kcal/mol. G3^{HC} is a subset of 46 G3 hydrocarbons, to which we have applied empirical corrections for the C atom as described in the text to show how consistently SCAN describes molecules. For all data sets, zero-point vibration effects have been removed from the reference experimental values. The LSDA results for G3 are from Ref. [25]. Becke-Lee-Yang-Parr (BLYP) [15,76], PBEsol [18], and PBE [6] are GGAs; SCAN, TPSS [7], and M06 L [20] are meta GGAs. We could not locate BLYP in VASP, but Ref. [77] suggests that its LC20 MAE may be more than twice that of PBE. (1 kcal/mol = 0.0434 eV.)

Regularized SCAN (rSCAN)



α is the iso-orbital indicator function

f_x , f'_x , f''_x are the switching functions

rSCAN has better numerical behavior than SCAN, but it is less accurate than SCAN

SCAN vs rSCAN vs r²SCAN

Table 2. Mean Error (ME) and Mean Absolute Error (MAE) of TPSS,⁵⁹ SCAN,⁷ rSCAN,²² and r²SCAN for the G3 Set of 226 Molecular Atomization Energies,²⁸ the BH76 Set of 76 Chemical Barrier Heights,⁵⁴ the S22 set of 22 Interaction Energies between Closed Shell Complexes,⁵⁵ and the LC20 Set of 20 Solid Lattice Constants^{56a}

	G3		BH76		S22		LC20	
	ME	MAE	ME	MAE	ME	MAE	ME	MAE
TPSS	−5.2	5.8	−8.6	8.6	−3.4	3.4	0.033	0.041
SCAN	−5.0	6.1	−7.7	7.7	−0.5	0.8	0.009	0.015
rSCAN	−14.0	14.3	−7.4	7.4	−1.2	1.3	0.020	0.025
r ² SCAN	−4.5	5.5	−7.1	7.2	−0.9	1.1	0.022	0.027

^aErrors for G3, BH76, and S22 sets are in kcal/mol, whereas errors for LC20 are in Å. We did not make corrections for basis set superposition error for the S22 set which used the aug-cc-pVTZ basis set.⁶⁰ All calculations for G3 and BH76 used the 6-311++G(3df,3pd) basis set.^{28,61} Details of the computational methods are included in Section S1 of the [Supporting Information](#).

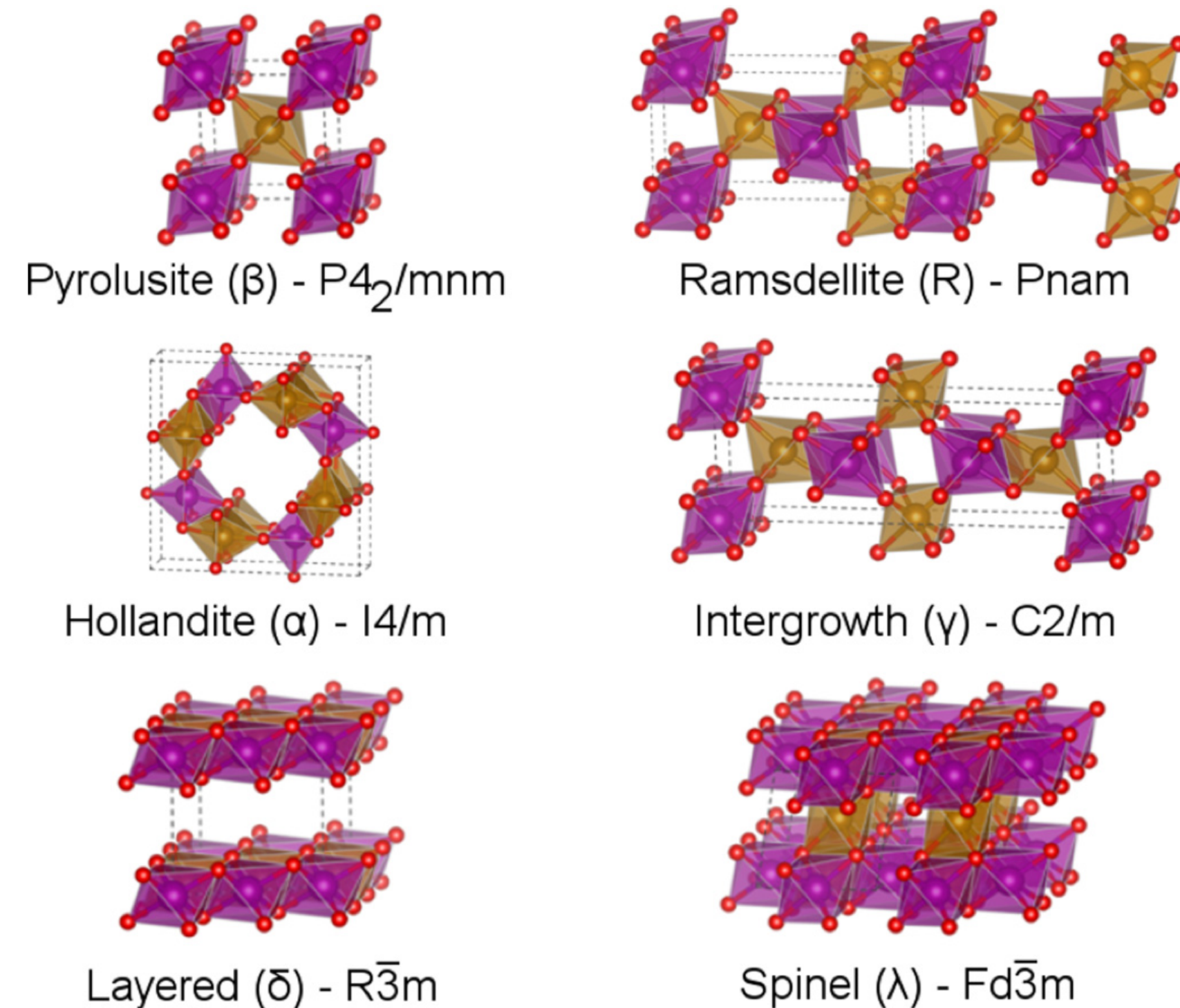
Success stories of SCAN

Material	Expt.	LDA	PBE	HSE	SCAN
Si	1.17	0.60	0.71	1.11	0.97
InP	1.42	0.50	0.72	1.52	1.06
GaAs	1.52	0.30	0.53	1.41	0.8
BAs	1.60 ^b	1.21	1.26	1.71	1.51
CdSe	1.73	0.44	0.71	1.66	1.10
BP	2.10	1.36	1.43	1.79	1.74
GaP	2.35	1.53	1.69	2.09	1.94
CdS	2.48	0.96	1.23	2.27	1.62
β -GaN	3.17	1.70	1.69	2.97	2.03
ZnS	3.72	1.87	2.12	3.32	2.63
C ^d	5.50	4.14	4.17	4.94	4.58

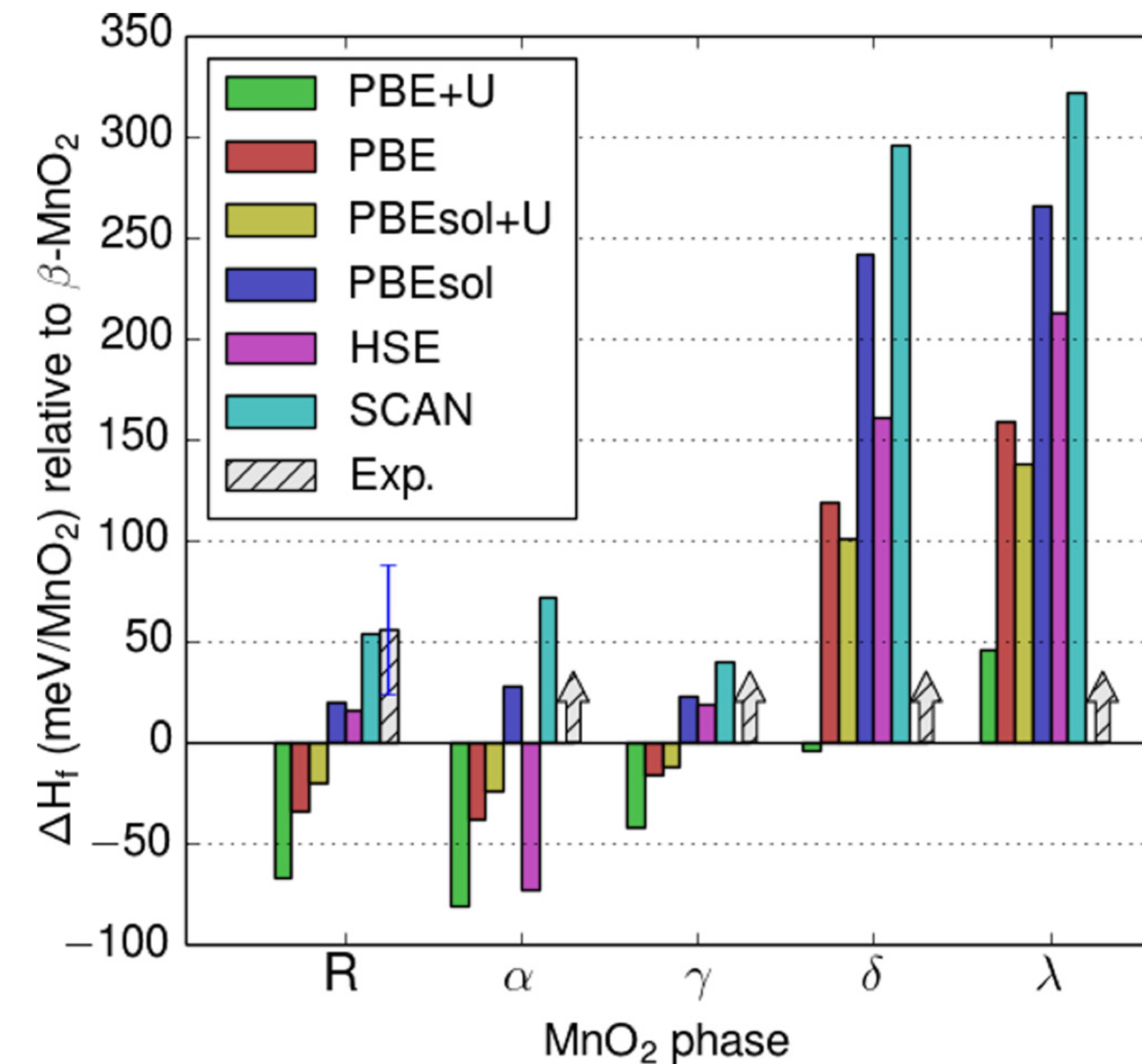
SCAN provides more accurate band gaps than LDA and PBE, but less accurate than HSE

Success stories of SCAN

Six polymorphs of MnO₂



Formation energies relative to the β phase



SCAN correctly predicts that the β phase is the lowest-energy phase

One of failures of SCAN

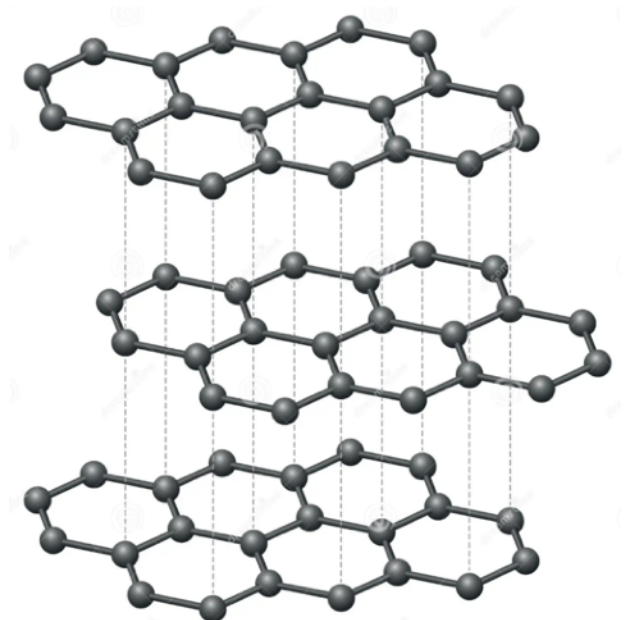
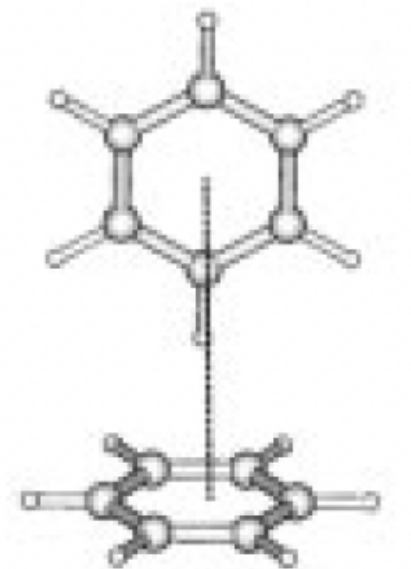
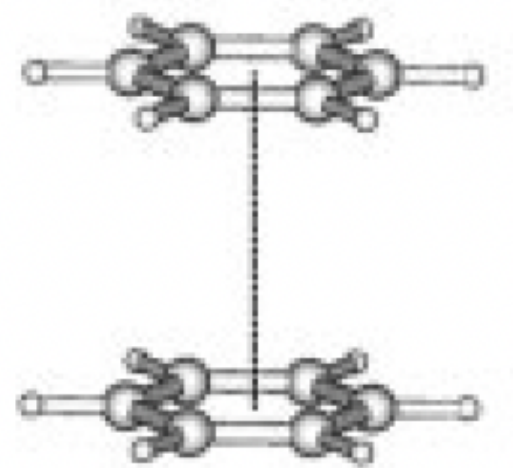
		V_0 (Å ³ /at)	B_0 (GPa)	B'_0 (1)	$m_s(V_0)$ (μ _B)
bcc-Fe	SCAN	11.58	157.5	5.05	2.66
	PBE	11.35	197.7	4.45	2.20
	LSDA	10.36	253.3	4.39	1.95
	Expt.	11.64	175.1	4.6	1.98, ^b 2.08, ^c 2.13 ^a
fcc-Ni	SCAN	10.38	230.5	4.79	0.73
	PBE	10.90	199.8	4.76	0.63
	LSDA	10.06	253.6	4.77	0.58
	Expt.	10.81	192.5	4	0.52, ^c 0.55, ^d 0.57 ^a
hcp-Co	SCAN	10.45	262.5	4.15	1.73
	PBE	10.91	196.9	4.61	1.61
	LSDA	9.99	237.6	4.95	1.49
	Expt.	10.96	198.4	4.26	1.52, ^c 1.55, ^b 1.58 ^e

SCAN overestimates magnetic moments in itinerant ferromagnets

van der Waals functionals

van der Waals (vdW) force

- vdW is a distance-dependent interaction between atoms or molecules
- vdW is weak, and it is different from ionic and covalent bonding
- vdW originate from correlations between charge fluctuations in different parts of an extended system (different fragments)
- vdW interactions are non-local
- vdW are not captured by LDA, GGA, meta-GGA (except SCAN for short-range part of vdW), DFT+U, and hybrids.



Two types of vdW functionals

vdW functionals

```
graph TD; A[vdW functionals] --> B[Empirical (or ab initio) corrections]; A --> C[Fully nonlocal functionals];
```

The diagram illustrates the classification of van der Waals (vdW) functionals. At the top, a red box labeled 'vdW functionals' has two arrows pointing down to two blue boxes. The left blue box is labeled 'Empirical (or *ab initio*) corrections' and contains the formula $C_6 R^{-6}$. The right blue box is labeled 'Fully nonlocal functionals' and contains the formula $\frac{1}{2} \int \int n(\mathbf{r}) \Phi(\mathbf{r}, \mathbf{r}') n(\mathbf{r}') d\mathbf{r} d\mathbf{r}'$.

Empirical (or *ab initio*) corrections

$$C_6 R^{-6}$$

Fully nonlocal functionals

$$\frac{1}{2} \int \int n(\mathbf{r}) \Phi(\mathbf{r}, \mathbf{r}') n(\mathbf{r}') d\mathbf{r} d\mathbf{r}'$$

Two types of vdW functionals

vdW functionals

```
graph TD; A[vdW functionals] --> B[Empirical (or ab initio) corrections]; A --> C[Fully nonlocal functionals];
```

The diagram illustrates the classification of van der Waals (vdW) functionals. At the top, a red box labeled 'vdW functionals' has two arrows pointing downwards to two separate boxes. The left box is blue and labeled 'Empirical (or *ab initio*) corrections', containing the formula $C_6 R^{-6}$. The right box is light blue and labeled 'Fully nonlocal functionals', containing the formula $\frac{1}{2} \iint n(\mathbf{r}) \Phi(\mathbf{r}, \mathbf{r}') n(\mathbf{r}') d\mathbf{r} d\mathbf{r}'$.

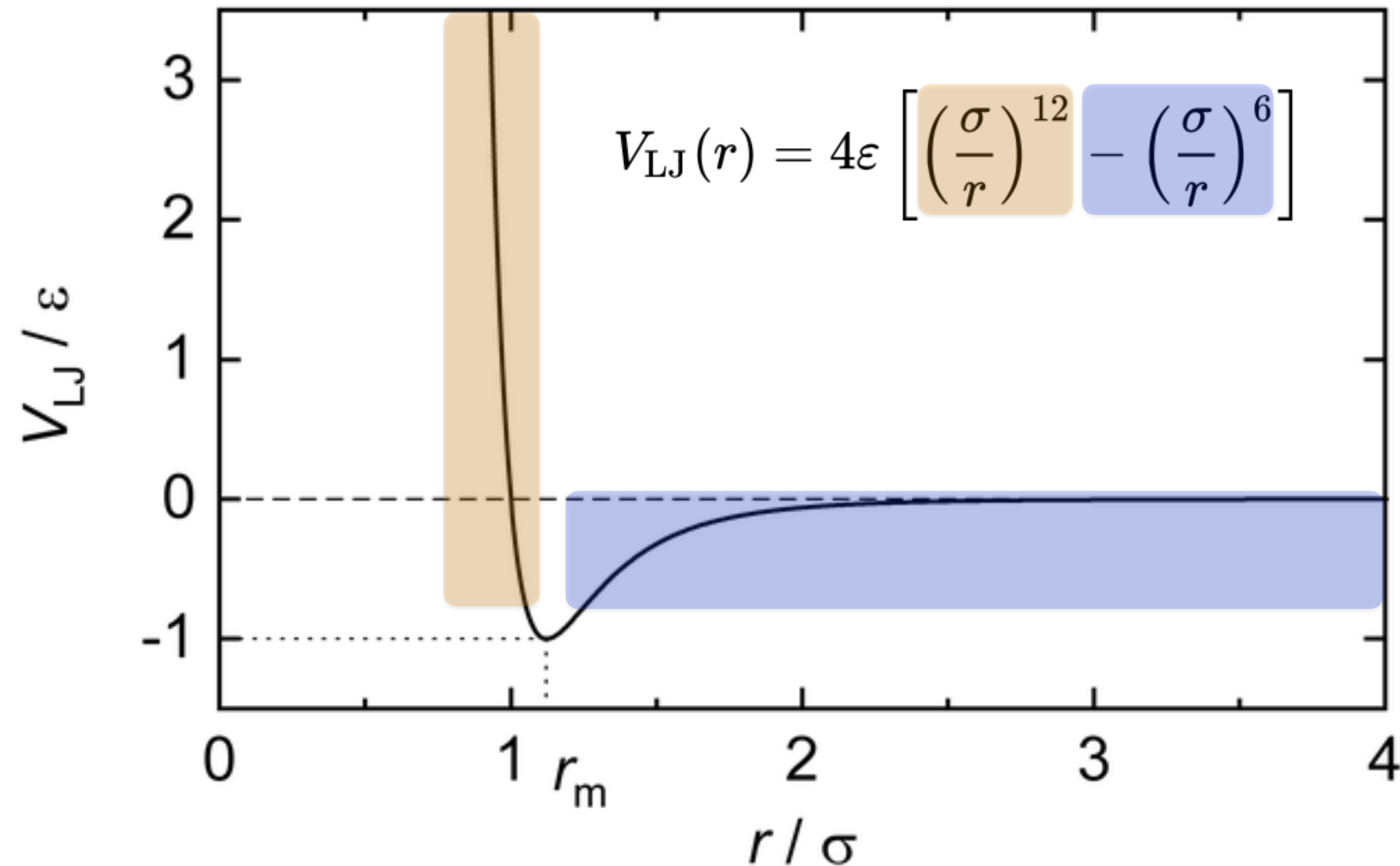
Empirical (or *ab initio*) corrections

$$C_6 R^{-6}$$

Fully nonlocal functionals

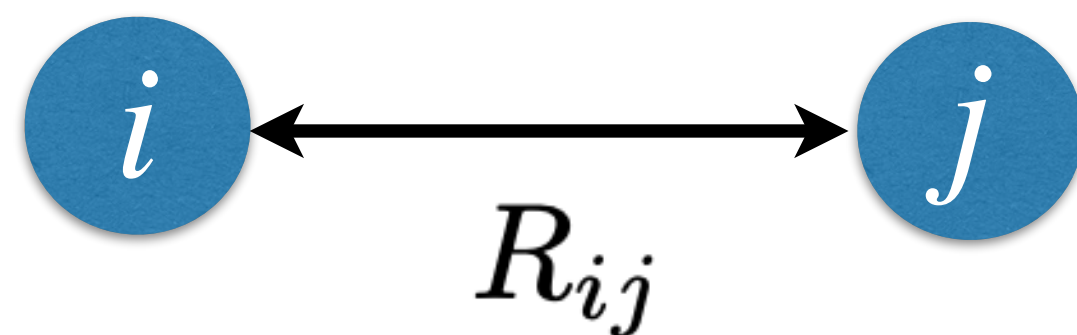
$$\frac{1}{2} \iint n(\mathbf{r}) \Phi(\mathbf{r}, \mathbf{r}') n(\mathbf{r}') d\mathbf{r} d\mathbf{r}'$$

Lennard-Jones potential



DFT-D functional

$$E_{\text{DFT-D}} = E_{\text{DFT}} + E_{\text{vdW}}, \quad E_{\text{vdW}} = -s_6 \sum_{i=1}^{N_{\text{at}}-1} \sum_{j=i+1}^{N_{\text{at}}} \frac{C_6^{ij}}{R_{ij}^6} f_{\text{dmp}}(R_{ij}), \quad f_{\text{dmp}}(R_{ij}) = \frac{1}{1 + e^{-d(R_{ij}/R_r - 1)}}$$



- Dispersion coefficients C_6^{ij} are determined empirically
- f_{dmp} is a damping function to avoid near-singularities for small R_{ij}

Element	C_6	R_0	Element	C_6	R_0
H	0.14	1.001	K	10.80 ^c	1.485
He	0.08	1.012	Ca	10.80 ^c	1.474
Li	1.61	0.825	Sc–Zn	10.80 ^c	1.562 ^d
Be	1.61	1.408	Ga	16.99	1.650
B	3.13	1.485	Ge	17.10	1.727
C	1.75	1.452	As	16.37	1.760
N	1.23	1.397	Se	12.64	1.771
O	0.70	1.342	Br	12.47	1.749
F	0.75	1.287	Kr	12.01	1.727
Ne	0.63	1.243	Rb	24.67 ^c	1.628
Na	5.71 ^c	1.144	Sr	24.67 ^c	1.606
Mg	5.71 ^c	1.364	Y–Cd	24.67 ^c	1.639 ^d
Al	10.79	1.639	In	37.32	1.672
Si	9.23	1.716	Sn	38.71	1.804
P	7.84	1.705	Sb	38.44	1.881
S	5.57	1.683	Te	31.74	1.892
Cl	5.07	1.639	I	31.50	1.892
Ar	4.61	1.595	Xe	29.99	1.881

DFT-D3 functional

$$E_{\text{DFT-D3}} = E_{\text{KS-DFT}} - E_{\text{vdW}}$$

$$E_{\text{vdW}} = E^{(2)} + E^{(3)}$$

DFT-D3 functional

$$E_{\text{DFT-D3}} = E_{\text{KS-DFT}} - E_{\text{vdW}}$$

$$E_{\text{vdW}} = E^{(2)} + E^{(3)}$$

Two-body term:

$$E^{(2)} = \sum_{AB} \sum_{n=6,8,10,\dots} s_n \frac{C_n^{AB}}{r_{AB}^n} f_{d,n}(r_{AB})$$

$$f_{d,n}(r_{AB}) = \frac{1}{1 + 6(r_{AB}/(s_{r,n}R_0^{AB}))^{-\alpha_n}}$$

DFT-D3 functional

$$E_{\text{DFT-D3}} = E_{\text{KS-DFT}} - E_{\text{vdW}}$$

$$E_{\text{vdW}} = E^{(2)} + E^{(3)}$$

Two-body term:

$$E^{(2)} = \sum_{AB} \sum_{n=6,8,10,\dots} s_n \frac{C_n^{AB}}{r_{AB}^n} f_{d,n}(r_{AB})$$

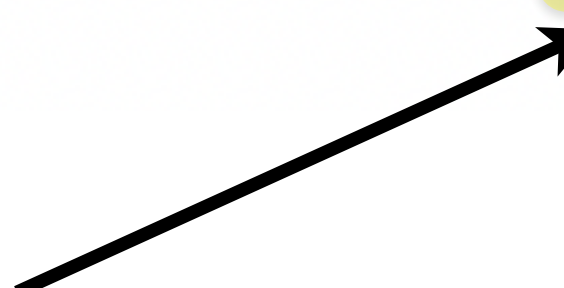
$$f_{d,n}(r_{AB}) = \frac{1}{1 + 6(r_{AB}/(s_n R_0^{AB}))^{-\alpha_n}}$$

Three-body term:

$$E^{(3)} = \sum_{ABC} f_{d,(3)}(\bar{r}_{ABC}) E^{ABC}$$

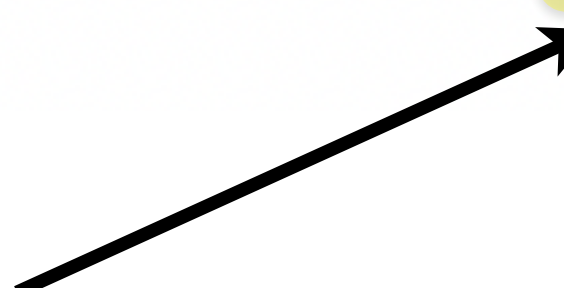
$$E^{ABC} = \frac{C_9^{ABC} (3 \cos \theta_a \cos \theta_b \cos \theta_c + 1)}{(r_{AB} r_{BC} r_{CA})^3}$$

Tkatchenko-Scheffler (TS) functional

$$E_{\text{vdW}} = -\frac{1}{2} \sum_{A,B} f_{\text{damp}}(R_{AB}, R_A^0, R_B^0) C_{6AB} R_{AB}^{-6}$$


Dispersion coefficients are computed from first principles using the electronic charge density

Tkatchenko-Scheffler (TS) functional

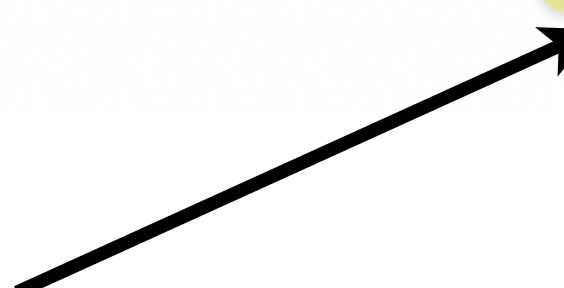
$$E_{\text{vdW}} = -\frac{1}{2} \sum_{A,B} f_{\text{damp}}(R_{AB}, R_A^0, R_B^0) C_{6AB} R_{AB}^{-6}$$


Dispersion coefficients are computed from first principles using the electronic charge density

$$C_{6AB} = \frac{2C_{6AA}C_{6BB}}{\left[\frac{\alpha_B^0}{\alpha_A^0} C_{6AA} + \frac{\alpha_A^0}{\alpha_B^0} C_{6BB}\right]}$$

$$C_{6AA}^{\text{eff}} = \frac{\eta_A^{\text{eff}}}{\eta_A^{\text{free}}} \left(\frac{\kappa_A^{\text{free}}}{\kappa_A^{\text{eff}}}\right)^2 \left(\frac{V_A^{\text{eff}}}{V_A^{\text{free}}}\right)^2 C_{6AA}^{\text{free}}$$

Tkatchenko-Scheffler (TS) functional

$$E_{\text{vdW}} = -\frac{1}{2} \sum_{A,B} f_{\text{damp}}(R_{AB}, R_A^0, R_B^0) C_{6AB} R_{AB}^{-6}$$


Dispersion coefficients are computed from first principles using the electronic charge density

$$C_{6AB} = \frac{2C_{6AA}C_{6BB}}{\left[\frac{\alpha_B^0}{\alpha_A^0} C_{6AA} + \frac{\alpha_A^0}{\alpha_B^0} C_{6BB}\right]}$$

$$C_{6AA}^{\text{eff}} = \frac{\eta_A^{\text{eff}}}{\eta_A^{\text{free}}} \left(\frac{\kappa_A^{\text{free}}}{\kappa_A^{\text{eff}}}\right)^2 \left(\frac{V_A^{\text{eff}}}{V_A^{\text{free}}}\right)^2 C_{6AA}^{\text{free}}$$

Hirshfeld partitioning of the electronic charge density:

$$\frac{\kappa_A^{\text{eff}}}{\kappa_A^{\text{free}}} \frac{\alpha_A^{\text{eff}}}{\alpha_A^{\text{free}}} = \frac{V_A^{\text{eff}}}{V_A^{\text{free}}} = \left(\frac{\int r^3 w_A(\mathbf{r}) n(\mathbf{r}) d^3\mathbf{r}}{\int r^3 n_A^{\text{free}}(\mathbf{r}) d^3\mathbf{r}} \right),$$

$$w_A(\mathbf{r}) = \frac{n_A^{\text{free}}(\mathbf{r})}{\sum_B n_B^{\text{free}}(\mathbf{r})},$$

Two types of vdW functionals

vdW functionals

```
graph TD; A[vdW functionals] --> B[Empirical (or ab initio) corrections]; A --> C[Fully nonlocal functionals];
```

The diagram illustrates the classification of van der Waals (vdW) functionals. At the top, a red box labeled 'vdW functionals' has two arrows pointing down to two separate boxes. The left box, labeled 'Empirical (or *ab initio*) corrections', is light blue and contains the formula $C_6 R^{-6}$. The right box, labeled 'Fully nonlocal functionals', is a darker blue and contains a double integral formula.

Empirical (or *ab initio*) corrections

$$C_6 R^{-6}$$

Fully nonlocal functionals

$$\frac{1}{2} \iint n(\mathbf{r}) \Phi(\mathbf{r}, \mathbf{r}') n(\mathbf{r}') d\mathbf{r} d\mathbf{r}'$$

Fully nonlocal vdW functionals

Nonlocal correlation energy: $E_c[n] = E_c^0[n] + E_c^{\text{nl}}[n]$

$E_c^{\text{nl}}[n]$ is defined to include the longest ranged or most nonlocal terms that give the vdW interaction and to approach zero in the limit of a slowly varying density. The term $E_c^0[n]$ is also nonlocal, but approaches the LDA in this limit.

$$E_c^{\text{nl}} = \frac{1}{2} \int \int n(\mathbf{r}) \Phi \left(n(\mathbf{r}), n(\mathbf{r}'), |\nabla n(\mathbf{r})|, |\nabla n(\mathbf{r}')|, |\mathbf{r} - \mathbf{r}'| \right) n(\mathbf{r}') d\mathbf{r} d\mathbf{r}'$$

nonlocal kernel

There are different numerical methods and approximations how to compute $E_c^{\text{nl}}[n]$

Fully nonlocal vdW functionals

vdW-DF

M. Dion et al., PRL 92, 246401 (2004).

vdW-DF2

K. Lee et al., PRB 82, 081101(R) (2010).

VV10

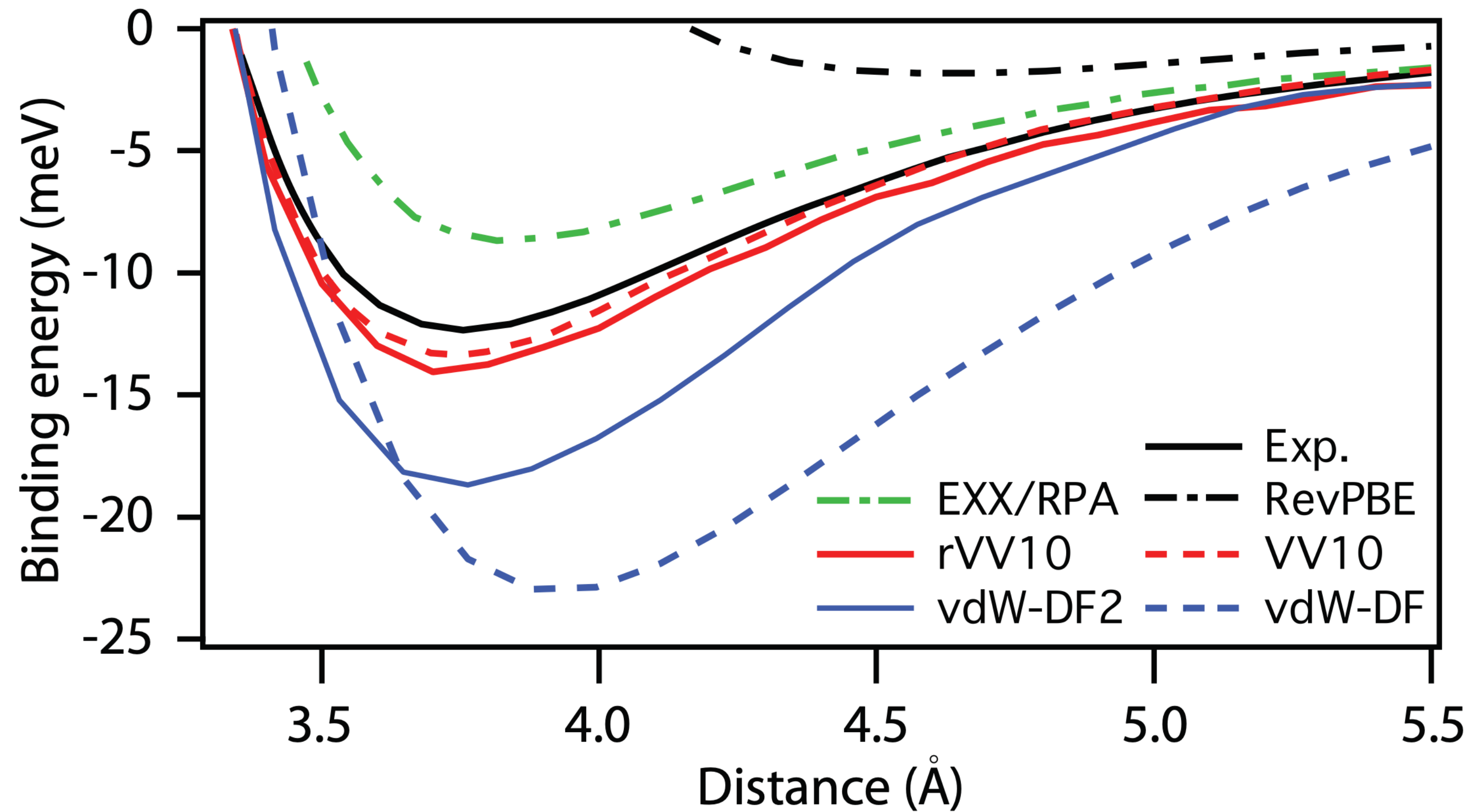
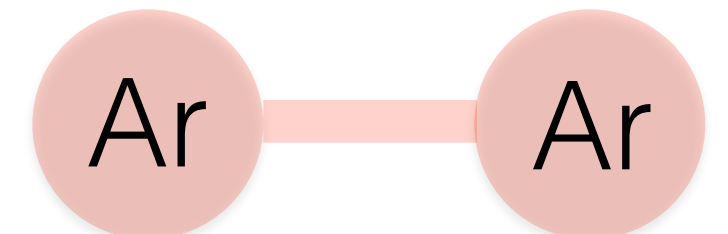
O.A. Vydrov and T. Van Voorhis, JCP 133, 244103 (2010).

rVV10

R. Sabatini, T. Gorni, S. de Gironcoli, PRB 87, 041108(R) (2013).

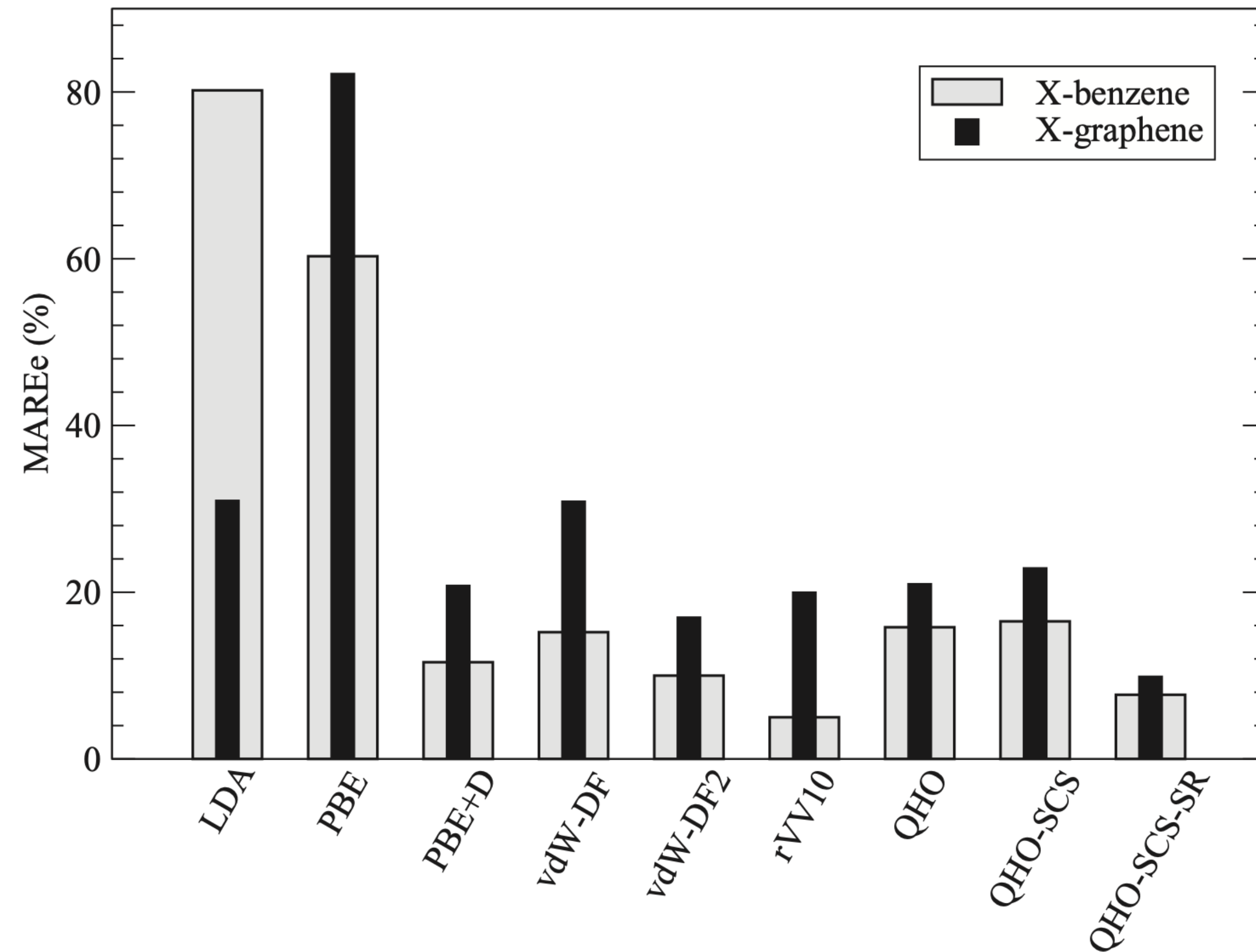
Examples

Binding energy curves for the Ar dimer



Examples

$X = \text{Ar}, \text{H}_2, \text{CO}, \text{H}_2\text{O}$



CO molecule

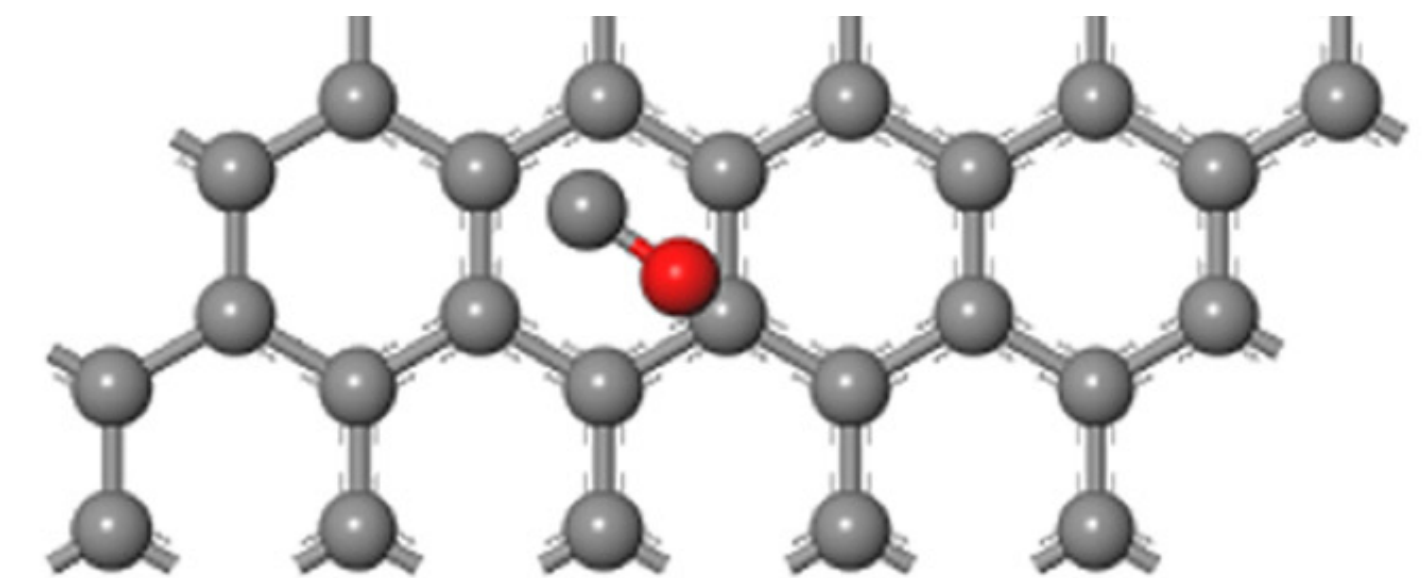


side view



graphene

top view



**Can we mix advanced
functionals?**

(meta-GGA) + (van der Waals) + (Hubbard U)

TABLE II. The calculated total energy with respect to the RS phase (ΔE in meV/f.u.), equilibrium volume (Ω_0 in $\text{\AA}^3/\text{f.u.}$), bulk modulus (B_0 in GPa), and fundamental band gap (E_g in eV) for MnO, FeO, CoO, and NiO in both RS and ZB structures, using PBE+TS+ U and SCAN+ r VV10+ U . The experimental data are collected in Refs. [17,18].

Compound		PBE+TS+ U				SCAN+ r VV10+ U				Experiment		
		ΔE	Ω_0	B_0	E_g	ΔE	Ω_0	B_0	E_g	Ω_0	B_0	E_g
MnO	RS	0	21.58	156	1.8	0	21.71	164	2.3	21.96	151–162	3.8–4.2
	ZB	88	26.47	102	0.8	138	26.43	120	1.1			
FeO	RS	0	20.25	150	1.2	0	20.28	163	1.4	20.35	150–180	2.4
	ZB	142	24.34	107	0.3	157	23.15	84	0.8			
CoO	RS	0	19.06	184	2.3	0	19.08	186	2.9	19.25	180	3.6
	ZB	51	23.53	137	1.4	92	23.45	146	1.7			
NiO	RS	0	17.76	205	2.5	0	17.94	220	3.5	18.14	166–208	3.7–4.3
	ZB	768	20.55	56	0.9	782	21.16	85	1.5			